

STAT 216 Coursepack



Fall 2026
Montana State University

Melinda Yager
Jade Schmidt
Dr. Mark Greenwood

This resource was developed by Melinda Yager, Jade Schmidt, and Stacey Hancock in 2021 to accompany the online textbook: Hancock, S., Carnegie, N., Meyer, E., Schmidt, J., and Yager, M. (2021). *Montana State Introductory Statistics with R*. Montana State University. <https://mtstateintrostats.github.io/IntroStatTextbook/>.

This resource is released under a Creative Commons BY-NC-SA 4.0 license unless otherwise noted.

Contents

Preface	1
1 Exploratory Data Analysis and Inference for Two Quantitative Variables	2
1.1 Vocabulary Review and Key Topics	2
1.2 Video Notes: Regression and Correlation	6
1.3 Activity 22: IPEDS (continued) — Linear Regression	24
1.4 Activity 20: Penguins	31
1.5 Activity 24: Golf Driving Distance	38
1.6 Module 13 Lab: Big Mac Index	47

Preface

This coursepack accompanies the textbook for STAT 216: Montana State Introductory Statistics with R, which can be found at <https://mtstateintrostats.github.io/IntroStatTextbook/>. The syllabus for the course (including the course calendar), data sets, and links to Canvas, Gradescope, and the MSU RStudio server can be found on the course webpage: <https://math.montana.edu/courses/s216/>. Other notes and review materials are linked in Canvas.

Each of the activities in this workbook is designed to target specific learning outcomes of the course, giving you practice with important statistical concepts in a group setting with instructor guidance. In addition to the in-class activities for the course, video notes are provided to aid in taking notes while you complete the required videos. Bring this workbook with you to class each class period, and take notes in the workbook as you would your own notes. A well-written completed workbook will provide an optimal study guide for exams!

All activities and labs in this coursepack will be completed during class time. Parts of each lab will be turned in on Gradescope. To aid in your understanding, read through the introduction for each activity before attending class each day.

STAT 216 is a 3-credit in-person course. In our experience, it takes six to nine hours per week outside of class to achieve a good grade in this class. By “good” we mean at least a C because a grade of D or below does not count toward fulfilling degree requirements. Many of you set your goals higher than just getting a C, and we fully support that. You need roughly nine hours per week to review past activities, read feedback on previous assignments, complete current assignments, and prepare for the next day’s class. A typical week in the life of a STAT 216 student looks like:

- *Prior to class meeting:*
 - Read assigned sections of the textbook, using the provided reading guides to take notes on the material.
 - Watch the provided videos, taking notes in the coursepack.
 - Read through the introduction to the day’s in-class activity.
 - Read through the week’s homework assignment and note any questions you may have on the content.
- *During class meeting:*
 - Work through the activities and labs with your classmates and instructor, taking detailed notes on your answers to each question in the activity.
- *After class meeting:*
 - Complete any parts of the activity you did not complete in class.
 - Review the activity solutions in the Math and Stat Center, and take notes on key points.
 - Complete any remaining assigned readings videos, and quizzes for the week.
 - Complete the week’s homework assignment.

Exploratory Data Analysis and Inference for Two Quantitative Variables

1.1 Vocabulary Review and Key Topics

Review the Golden Ticket posted in the resources at the end of the coursepack for a summary of two quantitative variables.

1.1.1 Key topics

Module 13 will cover exploratory data analysis and both simulation-based and theory-based methods of inference for two quantitative variables. The **summary measure** for two quantitative variables is either the **slope** of a regression line or the **correlation** between the two variables.

- Notation for a sample regression slope: b_1
- Notation for a population regression slope: β_1
- Notation for a sample correlation: r
- Notation for a population correlation: ρ

Types of plots for two quantitative variables:

- Scatterplot

1.1.2 Vocabulary

Plotting two quantitative variables

- **Scatterplot**: plots (x, y) pairs of observations with the explanatory variable on the x -axis and the response variable on the y -axis. R code to create a scatterplot:

```
object %>% # Pipe data set into...
ggplot(aes(x = explanatory, y = response)) + # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "x-axis label", # Label x-axis
       y = "y-axis label", # Label y-axis
       title = "Don't forget to add a title!") +
  # Be sure to tile your plots
  geom_smooth(method = "lm", se = FALSE) # Add regression line
```

- If there is a third categorical variable, you can use color or shape to include the third variable on the scatterplot.
- Four characteristics of scatterplots:
 - Form (linear or non-linear)
 - Direction (positive or negative)
 - Strength (weak, moderate, or strong)

- Outliers?

Sample statistics for two quantitative variables

- **Least-squares regression line:** a line fit to the data which minimizes the squared vertical distances from the observed y -value to the line

- **Notation for the fitted least-squares regression line:**

$$\hat{y} = b_0 + b_1 \times x$$

or

$$\widehat{response} = b_0 + b_1 \times explanatory$$

To write the equation of the regression line in context of the problem, include descriptive names of the response and explanatory variables for “ y /response” and “ x /explanatory” above.

- b_0 is the **y -intercept** of the regression line: the *predicted* value of the response variable when the explanatory variable is equal to zero.
- b_1 is the **slope** of the regression line: the *predicted* increase/decrease in the response variable associated with a one-unit increase in the explanatory variable.
- The distance from an observation’s y -value (observed response) to its fitted value, $\hat{y}$ (the value on the line) is called a **residual**:

$$residual = observed - fitted = y - \hat{y}$$

- A least-squares regression line is a special case of a **linear model**.
- R code to find the least-squares regression line (fit the linear model):

```
linearmodel <- lm(response~explanatory, data=object)
round(summary(linearmodel)$coefficients,3) # Display coefficients
```

- **Correlation:** measures the magnitude and direction of the linear relationship between two quantitative variables.

- Parameter notation: ρ
- Sample notation: r
- R code to find the **correlation** matrix between variables:

```
object %>% # Data set pipes into
  select(c("variable1", "variable2", "variable3")) %>%
  #Selects the variables you want to compare
  cor(use="pairwise.complete.obs") %>% #Calculates the correlation between each pair
  round(3) #Rounds to 3 decimal places
```

- **Coefficient of determination:** measures the proportion of total variability in the response variable that is explained by the linear relationship with the explanatory variable. The coefficient of determination can be calculated in three ways:

$$r^2 = (r)^2 = \frac{SST - SSE}{SST} = \frac{s_y^2 - s_{residual}^2}{s_y^2}$$

Hypotheses

Hypotheses involving two quantitative variables can be expressed either in terms of the slope or the correlation. When either the slope or correlation is equal to zero, there is no linear relationship between the two quantitative variables (the null hypothesis).

- Hypotheses in notation for slope:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \left\{ \begin{array}{l} < \\ \neq \\ < \end{array} \right\} 0$$

Simulation-based inference for two quantitative variables

- **Conditions necessary to use simulation-based methods for inference for two quantitative variables:**
 - **Independence:** observational units (the (x, y) pairs) must be independent of one another.
 - **Linearity:** the form of the relationship (if any) between the two variables must be linear.
- **Simulation-based methods to create the null distribution:** R code for simulation methods to find the p-value using the `regression_test` function in the `catstats` package.

```
regression_test(response~explanatory, # response ~ explanatory
  data = object, # Name of data set
  direction = "xx", # Sign in alternative ("greater", "less", "two-sided")
  summary_measure = "xx", # "slope" or "correlation"
  as_extreme_as = xx, # Observed slope or correlation
  number_repetitions = 10000) # Number of simulated samples for null distribution
```

- **Simulation-based methods to create the bootstrap distribution:** R code to find the simulation-based confidence interval using the `regression_bootstrap_CI` function from the `catstats` package.

```
regression_bootstrap_CI(response~explanatory, # response ~ explanatory
  data = object, # Name of data set
  confidence_level = xx, # Confidence level as decimal
  summary_measure = "xx", # Slope or correlation
  number_repetitions = 10000) # Number of simulated samples for bootstrap distribution
```

Theory-based methods for two quantitative variables

- **Conditions necessary to use theory-based methods for inference for two quantitative variables:**
 - **Independence** (for both simulation-based and theory-based methods): observational units (the (x, y) pairs) must be independent of one another.
 - * Check this assumption by investigating the sampling method and determining if the observational units are related in any way.
 - **Linearity** (for both simulation-based and theory-based methods): the form of the relationship (if any) between the two variables must be linear.
 - * Check this assumption by examining the scatterplot of the two variables, and a scatterplot of the residuals (on the y -axis) versus the fitted values (on the x -axis). The pattern in the residuals vs. fitted plot should display a horizontal line.
 - **Constant variability** (for theory-based methods only): the variability of points around the least squares line remains roughly constant

- * Check this assumption by examining a scatterplot of the residuals (on the y -axis) versus the fitted values (on the x -axis). The variability in the residuals around zero should be approximately the same for all fitted values.
- **Nearly normal residuals** (for theory-based methods only): residuals must be nearly normal.
 - * Check this assumption by examining a histogram of the residuals, which should appear approximately normal.
- **Standard error of the slope of the least-squares regression line** ($SE(b_1)$): obtain the value of the standard error of the slope from the linear model (`lm`) R output.
- **Standardized slope:**

$$T = \frac{\text{slope estimate} - \text{nullvalue}}{SE} = \frac{b_1 - 0}{SE(b_1)}.$$
 - The p-value can be found from the linear model (`lm`) R output or by using the `pt` function in R to find the area under a t -distribution with $n - 2$ degrees of freedom where T is as or more extreme as the value observed (in the direction of H_A).
- **Margin of error:** half the width of the confidence interval. For a regression slope, the margin of error is:

$$ME = t^* \times SE(b_1)$$

where t^* is the **multiplier**, corresponding to the desired confidence level found from a t -distribution with $n - 2$ degrees of freedom.

- Use the `qt` function in R to find the t^* multiplier with $n - 2$ degrees of freedom.
- To find the endpoints of a confidence interval, add and subtract the margin of error to the sample statistic. The confidence interval for a population slope is:

$$b_1 \pm ME$$

1.2 Video Notes: Regression and Correlation

Read Chapters 6, 7, 8, 21, and 22 in the course textbook. Use the following videos to complete the video notes for Module 12.

1.2.1 Course Videos

- 6.1
- 6.2
- 6.3
- Ch 7
- 21.4TheoryTests
- 21.4TheoryIntervals
- Optional: 21.1
- Optional: 21.3

Summary measures and plots for two quantitative variables - Videos 6.1 - 6.3

Example: Data were collected from 1236 births between 1960 and 1967 in the San Francisco East Bay area to better understand what variables contributed to child birthweight, as children with low birthweight often suffer from an array of complications later in life (*Child Health and Development Studies*, n.d.). There were some missing values in the study and with those observations removed we have a total of 1223 births.

```
babies<-read.csv("data/babies.csv") %>%
  drop_na(bwt) %>%
  drop_na(gestation)
glimpse(babies)
#> Rows: 1,223
#> Columns: 8
#> $ case      <int> 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, ~
#> $ bwt       <int> 120, 113, 128, 108, 136, 138, 132, 120, 143, 140, 144, 141, ~
#> $ gestation <int> 284, 282, 279, 282, 286, 244, 245, 289, 299, 351, 282, 279, ~
#> $ parity    <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
#> $ age       <int> 27, 33, 28, 23, 25, 33, 23, 25, 30, 27, 32, 23, 36, 30, 38, ~
#> $ height    <int> 62, 64, 64, 67, 62, 62, 65, 62, 66, 68, 64, 63, 61, 63, 63, ~
#> $ weight    <int> 100, 135, 115, 125, 93, 178, 140, 125, 136, 120, 124, 128, 9~
#> $ smoke     <int> 0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 1, 1, 0, 1, ~
```

Here you see a glimpse of the data. The 1223 rows correspond to the sample size. The case variable is labeling each pregnancy 1 through 1223. Then 7 variables are recorded. birthweight (bwt), length of gestation in days, parity is called an indicator variable telling us if the pregnancy was a first pregnancy (labeled as 0) or not (labeled as 1) were recorded about the child and pregnancy. The age, height, and weight were recorded for the mother giving birth, as was smoke, another indicator variable where 0 means the mother did not smoke during pregnancy, and 1 indicates that she did smoke while pregnant.

Type of plot

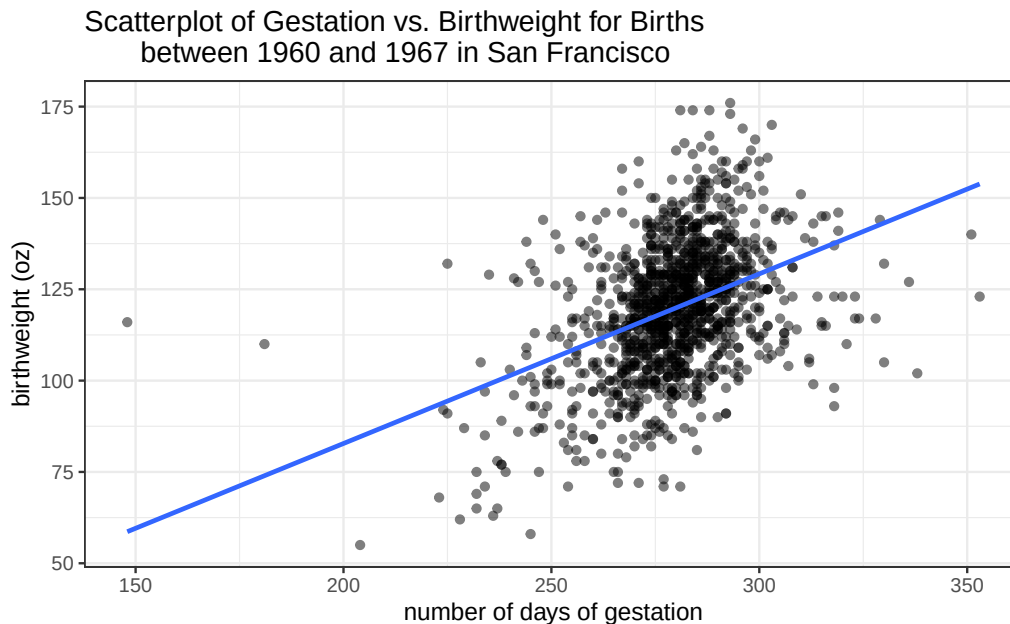
A _____ is used to display the relationship between two _____ variables.

Four characteristics of the scatterplot:

- Form:
- Direction:
- Strength:
- Outliers:
 - Influential points: outliers that change the regression line; far from the line of regression
 - High leverage points: outliers that are extreme in the x- axis; far from the mean of the x-axis

The following shows a scatterplot of length of gestation as a predictor of birthweight.

```
babies %>% # Data set pipes into...
ggplot(aes(x = gestation, y = bwt))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "number of days of gestation", # Label x-axis
       y = "birthweight (oz)", # Label y-axis
       title = "Scatterplot of Gestation vs. Birthweight for Births
               between 1960 and 1967 in San Francisco") +
  # Be sure to title your plots with the type of plot, observational units, variable(s)
  geom_smooth(method = "lm", se = FALSE) + # Add regression line
  theme_bw()
```



Describe the scatterplot using the four characteristics of a scatterplot.

The summary measures for two quantitative variables are:

- _____
- _____
- _____

Notation:

- Population slope:
- Population correlation:
- Sample slope:
- Sample correlation:

Correlation

Correlation is always between the values of _____ and _____.

- Measures the _____ and _____ of the linear relationship between two quantitative variables.
- The stronger the relationship between the variables the closer the value of _____ is to _____ or _____.
- The sign gives the _____.

The following code creates a correlation matrix between different quantitative variables in the data set.

```
babies %>%  
  select(c("gestation", "age", "height", "weight", "bwt")) %>%  
  cor(use="pairwise.complete.obs") %>%  
  round(3)
```

```
#>      gestation    age height weight    bwt  
#> gestation      1.000 -0.056  0.064  0.022 0.408  
#> age           -0.056  1.000 -0.005  0.147 0.029  
#> height        0.064 -0.005  1.000  0.436 0.201  
#> weight        0.022  0.147  0.436  1.000 0.154  
#> bwt           0.408  0.029  0.201  0.154 1.000
```

The value of correlation between gestation and birthweight is _____. This shows a _____, _____ relationship between gestation and birthweight.

Slope

- Least-squares regression line: $\hat{y} = b_0 + b_1 \times x$ (put y and x in the context of the problem) or $\widehat{response} = b_0 + b_1 \times \text{explanatory}$
- $\hat{y}$ or $\widehat{response}$ is
- b_0 is
- b_1 is
- x or explanatory is

- The estimates for the linear model output will give the value of the _____ and the _____.
- Interpretation of slope: an increase in the _____ variable of 1 unit is associated with an increase/decrease in the _____ variable by the value of slope, on average.
- Interpretation of the y-intercept: for a value of 0 for the _____ variable, the predicted value for the _____ variable would be the value of y-intercept.
- We can predict values of the _____ variable by plugging in a given _____ variable value using the least squares equation line.
- A prediction of a response variable value for an explanatory value outside the range of x values is called _____.
- To find how far the predicted value deviates from the actual value we find the _____.

- To find the least squares regression line the line with the _____ SSE is found.

SSE = sum of squared errors

- To find SSE, the residual for each data point is found, squared and all the squared residuals are summed together

The linear model output for this study is given below:

```
# Fit linear model: y ~ x
babiesLM <- lm(bwt ~ gestation, data=babies)
round(summary(babiesLM)$coefficients,3) # Display coefficient summary
```

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  -10.064      8.322   -1.209   0.227
#> gestation      0.464      0.030   15.609   0.000
```

Write the least squares equation of the line.

Interpret the slope in context of the problem.

Interpret the y-intercept in context of the problem.

Predict the birthweight for a birth with a baby born at 310 days gestation.

Calculate the residual for a birth of a baby with a birthweight of 151 ounces and born at 310 days gestation.

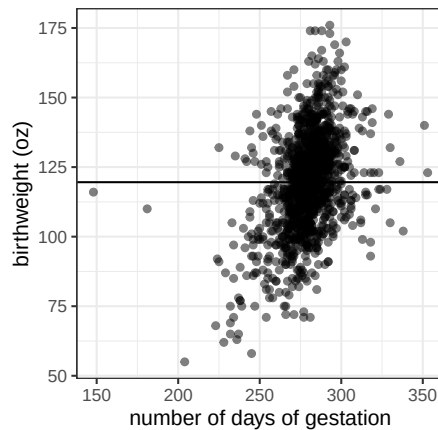
Is this value (310, 151) above or below the line of regression? Did the line of regression overestimate or underestimate the birthweight?

Coefficient of Determination

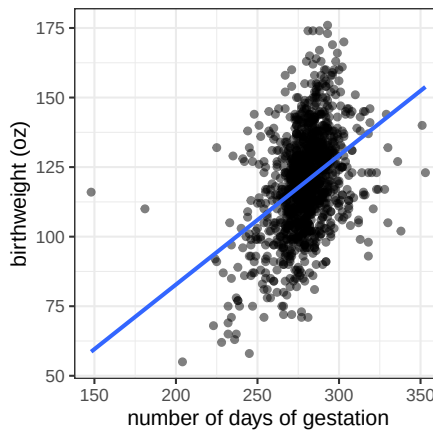
The coefficient of determination can be found by squaring the value of correlation, using the variances for each variable or using the SSE (sum of squares error) and SST (sum of squares total)

- $r^2 = (r)^2 = \frac{SST - SSE}{SST} = \frac{s_y^2 - s_{residual}^2}{s_y^2}$
- The coefficient of determination measures the _____ of total variation in the _____ variable that is explained by the changes in the _____ variable.

A Scatterplot of Gestation vs. Birthweight for Births between 1960 and 1967 in SF with Horizontal Line



B Scatterplot of Gestation vs. Birthweight for Births between 1960 and 1967 in SF with Regression Line



The value for SST was calculated as 406753.48. The value for SSE was calculated as 339092.13.

Calculate the coefficient of determination between gestation and birthweight.

Interpret the coefficient of determination between gestation and birthweight.

Multivariable plots - Video Chapter7

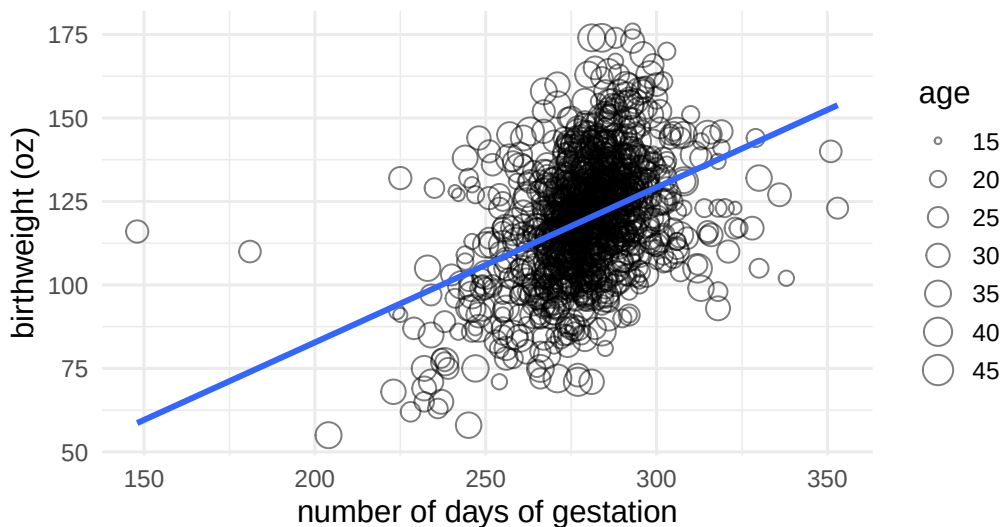
Aesthetics: visual property of the objects in your plot

- Position on the axes: groups for _____ variables, or a number line if the variable is _____
- Color or shape - to represent _____ variables
- Size - to represent _____ variables

Adding the quantitative variable maternal age to the scatterplot between gestation and birthweight.

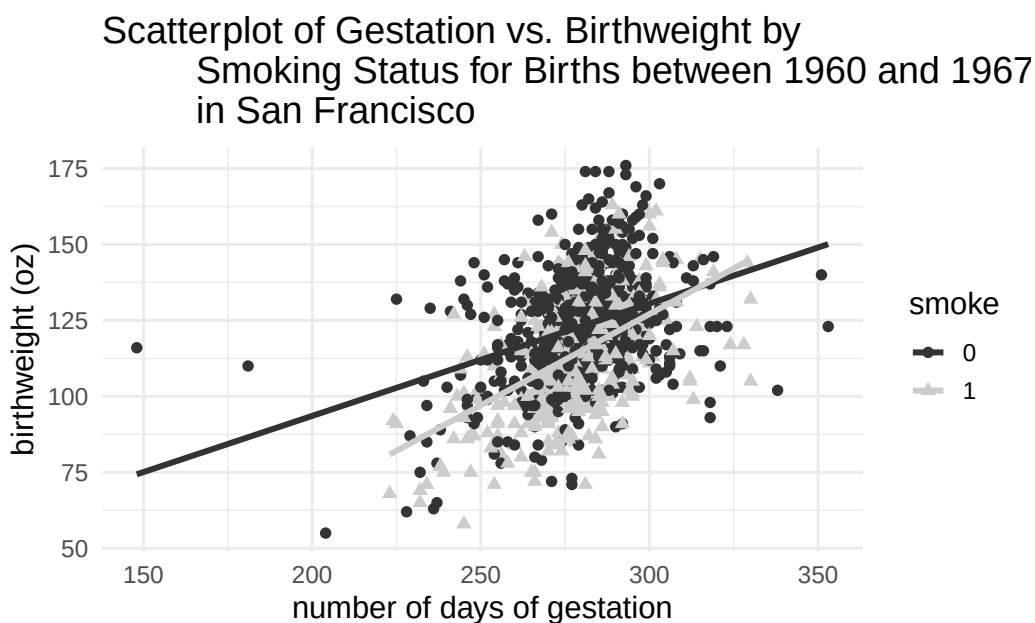
```
babies %>% # Data set pipes into...
ggplot(aes(x = gestation, y = bwt))+ # Specify variables
  geom_point(alpha=0.5, shape=1, aes(size=age)) + # Add scatterplot of points
  labs(x = "number of days of gestation", # Label x-axis
       y = "birthweight (oz)", # Label y-axis
       title = "Scatterplot of Gestation vs. Birthweight by Age
               for Births between 1960 and 1967 in San Francisco") +
  # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) # Add regression line
```

Scatterplot of Gestation vs. Birthweight by Age
for Births between 1960 and 1967 in San Francisco



Let's add the categorical variable, whether a mother smoked, to the scatterplot between gestation and birthweight.

```
babies <- babies %>%  
  mutate(smoke = factor(smoke)) %>%  
  na.omit()  
  
babies %>% # Data set pipes into...  
  ggplot(aes(x = gestation, y = bwt, color = smoke)) + #Specify variables  
  geom_point(aes(shape = smoke), size = 2) + #Add scatterplot of points  
  labs(x = "number of days of gestation", #Label x-axis  
       y = "birthweight (oz)", #Label y-axis  
       title = "Scatterplot of Gestation vs. Birthweight by  
               Smoking Status for Births between 1960 and 1967  
               in San Francisco") +  
  #Be sure to title your plots  
  geom_smooth(method = "lm", se = FALSE) + #Add regression line  
  scale_color_grey()
```



Does the relationship between length of gestation and birthweight appear to depend upon maternal smoking status?

Is the variable smoking status a potential confounding variable?

Adding a categorical predictor:

- Look at the regression line for each level of the _____
- If the slopes are _____, the two predictor variables do not _____ to help explain the response
- If the slopes _____, there is an interaction between the categorical predictor and the relationship between the two quantitative variables.

1.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. What are the three summary measures for two quantitative variables?
2. What are the four characteristics used to describe a scatterplot?
3. When we add a categorical predictor variable to a scatterplot of two quantitative variables, what summary measure will we compare across the categories to assess the change in the relationship between the two quantitative variables.

Theoretical Testing for Slope - Video 21.4to21.5TheoryTests

Conditions:

- Linearity (for both simulation-based and theory-based methods): the data should follow a linear trend.
 - Check this assumption by examining the _____ of the two variables, and _____. The pattern in the residual plot should display a horizontal line.
- Independence (for both simulation-based and theory-based methods)
 - One _____ for an observational unit has no impact on _____.
- Constant variability (for theory-based methods only): the variability of points around the least squares line remains roughly constant
 - Check this assumption by examining the _____. The variability in the residuals around zero should be approximately the same for all fitted values.
- Nearly normal residuals (for theory-based methods only): residuals must be nearly normal
 - Check this assumption by examining a _____, which should appear approximately normal

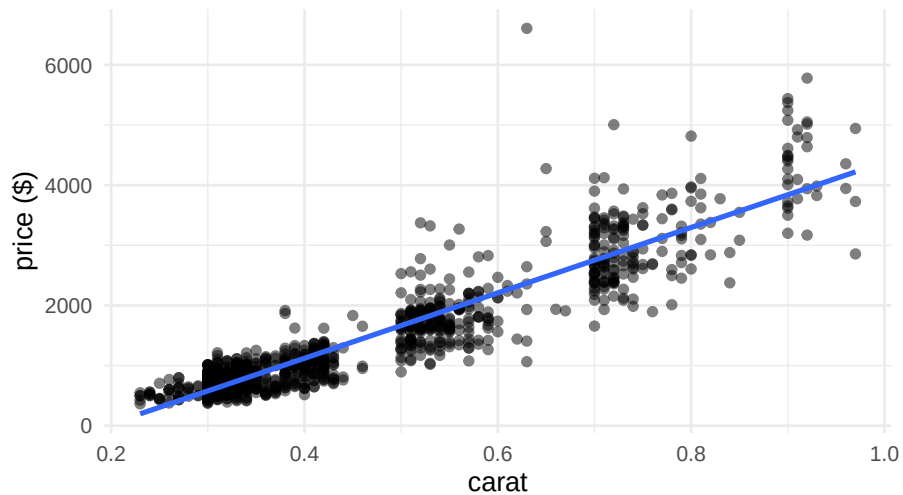
Example:

It is a generally accepted fact that the more carats a diamond has, the more expensive that diamond will be. The question is, how much more expensive? Data on thousands of diamonds were collected for this data set. We will only look at one type of cut ("Ideal") and diamonds less than 1 carat. Does the association between carat size and price have a linear relationship for these types of diamonds? What can we state about the association between carat size and price?

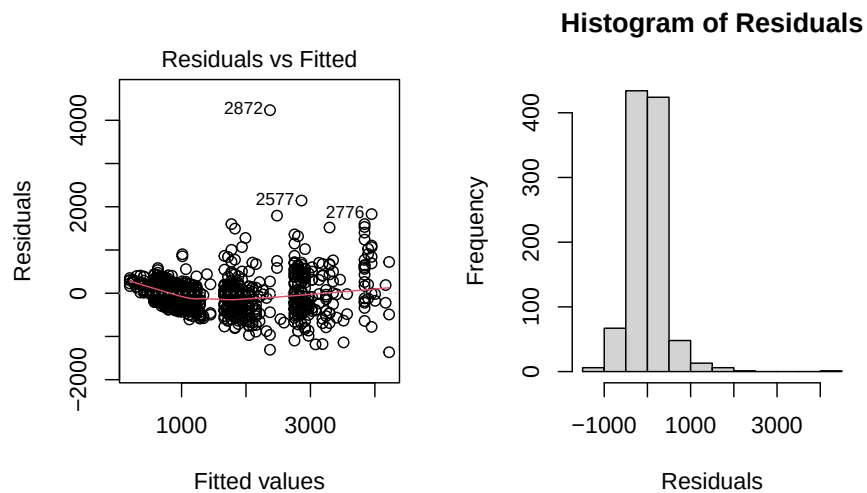
Scatterplot:

```
Diamonds %>% # Pipe data set into...
  ggplot(aes(x = carat, y = price))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "carat", # Label x-axis
       y = "price ($)", # Label y-axis
       title = "Scatterplot of Diamonds Carats vs Price") +
  # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) # Add regression line
```

Scatterplot of Diamonds Carats vs Price



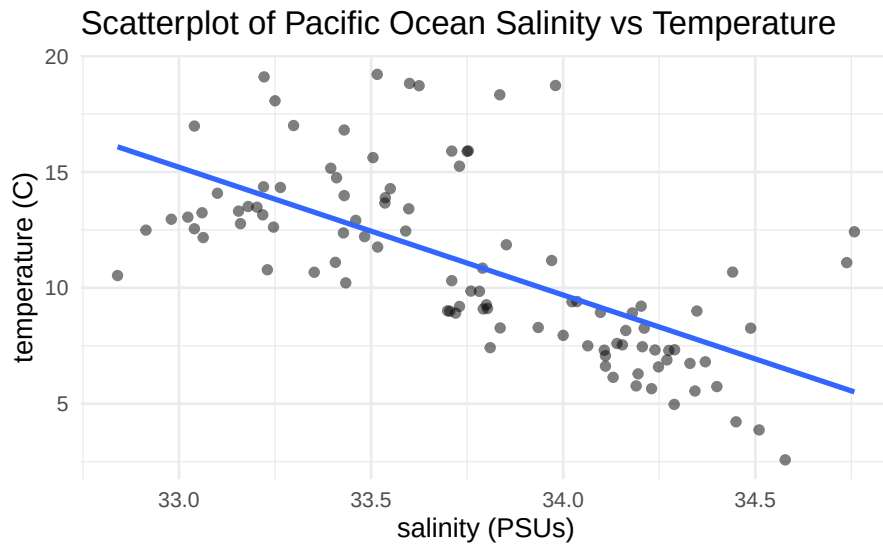
Diagnostic plots:



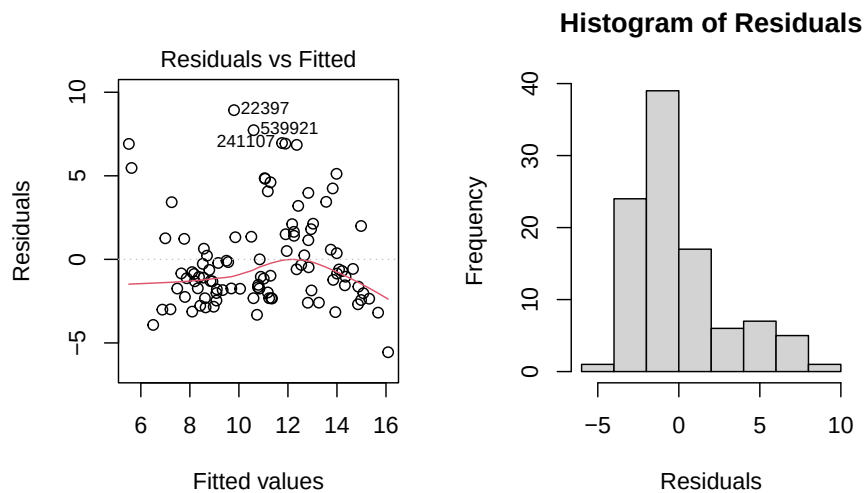
Check the conditions for the ocean data:

Scatterplot:

```
water %>% # Pipe data set into...
ggplot(aes(x = Salnty, y = T_degC))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "salinity (PSUs)", # Label x-axis
       y = "temperature (C)", # Label y-axis
       title = "Scatterplot of Pacific Ocean Salinity vs Temperature") +
  # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) # Add regression line
```



Diagnostic plots:



Like with paired data the t -distribution can be used to model slope and correlation.

- For two quantitative variables we use the _____-distribution with _____ degrees of freedom to approximate the sampling distribution.

Theory-based test:

- Calculate the standardized statistic
- Find the area under the t -distribution with $n - 2$ df at least as extreme as the standardized statistic

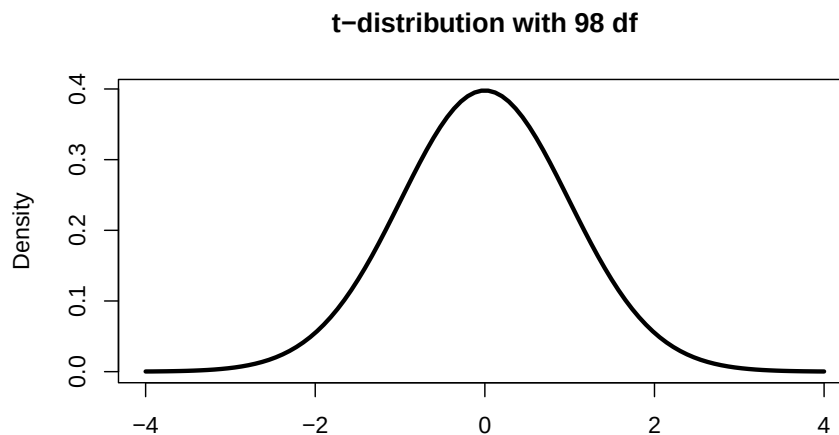
Equation for the standardized slope:

Optional Notes: Video Example (Video 21.4TheoryTests)

Calculate the standardized slope for the ocean data

```
lm.water <- lm(T_degC~Salnty, data=water) # lm(response-explanatory)
round(summary(lm.water)$coefficients,3)
```

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  197.156      21.478    9.18    0
#> Salnty       -5.514       0.636   -8.67    0
```



Interpret the standardized statistic:

To find the theory-based p-value:

```
lm.water <- lm(T_degC~Salnty, data=water) # lm(response-explanatory)
round(summary(lm.water)$coefficients,3)
```

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  197.156      21.478    9.18    0
#> Salnty       -5.514       0.636   -8.67    0
```

or

```
pt(-8.670, df = 98, lower.tail=TRUE)
#> [1] 4.623445e-14
```

Theoretical Confidence Interval for Slope - Video 21.4TheoryInterval

- Calculate the interval centered at the sample statistic
 $\text{statistic} \pm \text{margin of error}$

Optional Notes: Video Example (Video 21.4TheoryInterval)

```
lm.water <- lm(T_degC~Salnty, data=water) # lm(response~explanatory)
round(summary(lm.water)$coefficients, 3)
```

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   197.156      21.478    9.18    0
#> Salnty        -5.514       0.636   -8.67    0
```

Using the ocean data, calculate a 95% confidence interval for the true slope.

- Need the t^* multiplier for a 95% confidence interval from a t-distribution with _____ df.

```
qt(0.975, df=98, lower.tail = TRUE)
```

```
#> [1] 1.984467
```

Video Notes: Inference for Two Quantitative Variables

Example: Oceanic temperature is important for sea life. The California Cooperative Oceanic Fisheries Investigations has measured several variables on the Pacific Ocean for more than 70 years hoping to better understand weather patterns and impacts on ocean life. (*Ocean Temperature and Salinity Study*, n.d.) For this example, we will look at the most recent 100 measurements of salt water salinity (measured in PSUs or practical salinity units) and the temperature of the ocean measured in degrees Celsius. Is there evidence that water temperature in the Pacific Ocean tends to decrease with higher levels of salinity?

Hypothesis Testing - Video 21.1

Null hypothesis assumes “no effect”, “no difference”, “nothing interesting happening”, etc.

Always of form: “parameter” = null value

H_0 :

H_A :

- Research question determines the alternative hypothesis.

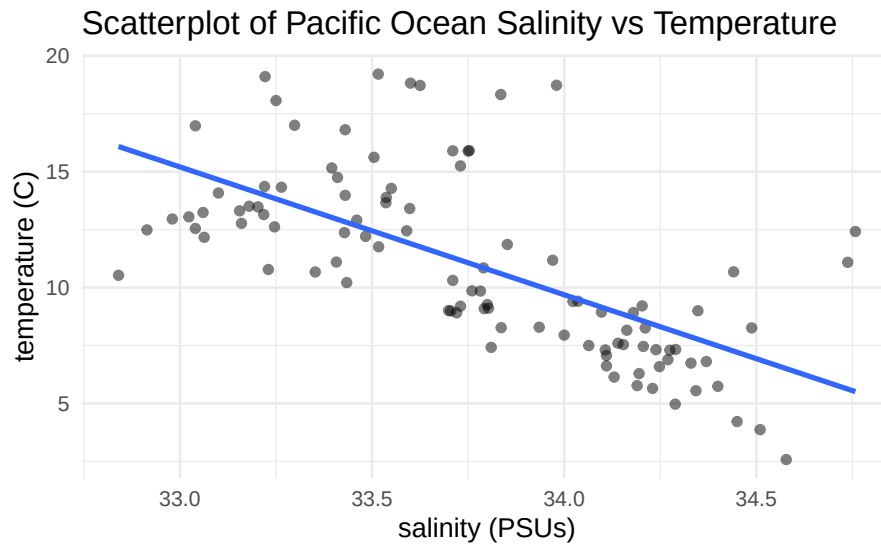
Write the null and alternative for the ocean study:

In notation:

H_0 :

H_A :

```
water %>% # Pipe data set into...
ggplot(aes(x = Salnty, y = T_degC))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "salinity (PSUs)", # Label x-axis
       y = "temperature (C)", # Label y-axis
       title = "Scatterplot of Pacific Ocean Salinity vs Temperature") +
  # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) # Add regression line
```



Simulation-based method

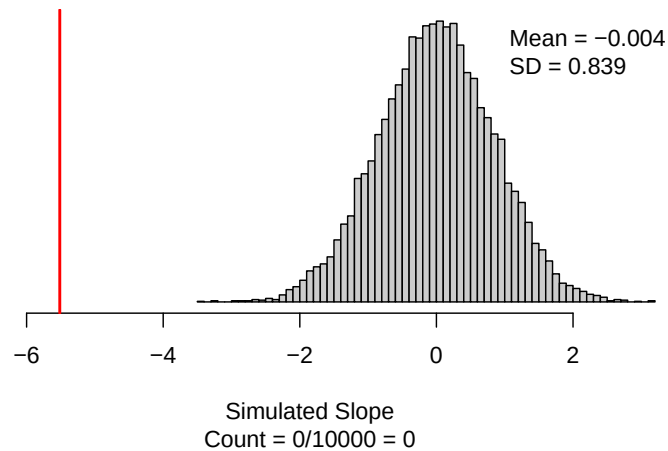
Conditions:

- Independence: the response for one observational unit will not influence another observational unit
- Linear relationship:

- Simulate many samples assuming $H_0 : \beta_1 = 0$ or $H_0 : \rho = 0$
 - Write the response variable values on cards
 - Hold the explanatory variable values constant
 - Shuffle a new response variable to an explanatory variable
 - Plot the shuffled data points to find the least squares line of regression
 - Calculate and plot the simulated slope or correlation from each simulation
 - Repeat 10000 times (simulations) to create the null distribution
 - Find the proportion of simulations at least as extreme as b_1 or r

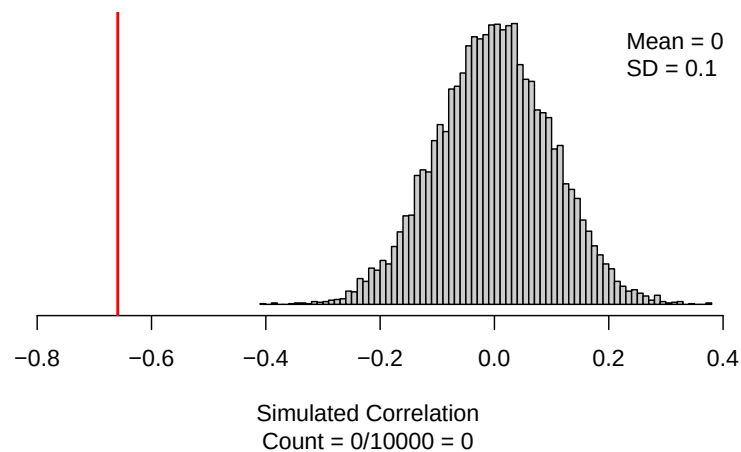
To test slope:

```
set.seed(216)
regression_test(T_degC ~ Salnty, # response ~ explanatory
  data = water, # Name of data set
  direction = "less", # Sign in alternative ("greater", "less", "two-sided")
  summary_measure = "slope", # "slope" or "correlation"
  as_extreme_as = -5.514, # Observed slope or correlation
  number_repetitions = 10000) # Number of simulated samples for null distribution
```



To test correlation:

```
set.seed(216)
regression_test(T_degC~Salnty, # response ~ explanatory
  data = water, # Name of data set
  direction = "less", # Sign in alternative ("greater", "less", "two-sided")
  summary_measure = "correlation", # "slope" or "correlation"
  as_extreme_as = -0.659, # Observed slope or correlation
  number_repetitions = 10000) # Number of simulated samples for null distribution
```



Explain why the null distribution is centered at the value of zero:

Interpretation of the p-value:

- Statement about probability or proportion of samples
- Statistic (summary measure and value)
- Direction of the alternative
- Null hypothesis (in context)

Conclusion:

- Amount of evidence
- Parameter of interest
- Direction of the alternative hypothesis

Confidence interval - Video 21.3

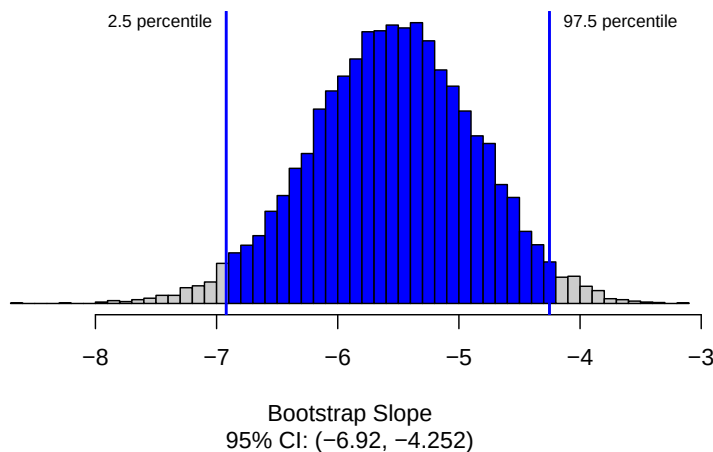
To estimate the true slope (or true correlation) we will create a confidence interval.

Simulation-based method

- Write the explanatory and response value pairs on cards
- Sample pairs with replacement n times
- Plot the resampled data points to find the least squares line of regression
- Calculate and plot the simulated slope (or correlation) from each simulation
- Repeat 10000 times (simulations) to create the bootstrap distribution
- Find the cut-offs for the middle X% (confidence level) in a bootstrap distribution.

Returning to the ocean example, we will estimate the true slope between salinity and temperature of the Pacific Ocean.

```
set.seed(216)
regression_bootstrap_CI(T_degC~Salnty, # response ~ explanatory
  data = water, # Name of data set
  confidence_level = 0.95, # Confidence level as decimal
  summary_measure = "slope", # Slope or correlation
  number_repetitions = 10000) # Number of simulated samples for bootstrap distribution
```

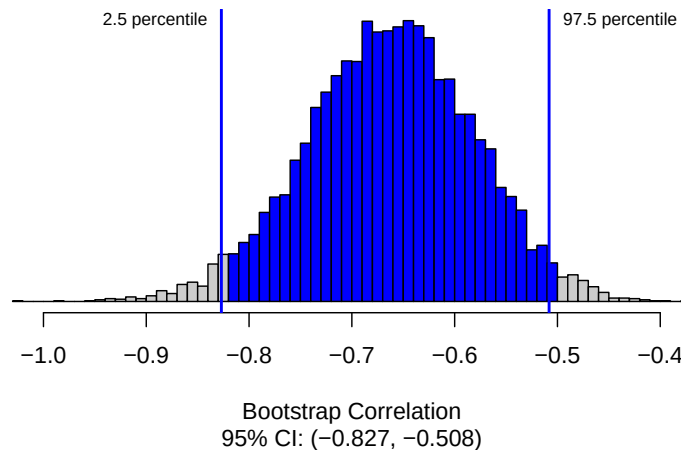


Confidence interval interpretation:

- How confident you are (e.g., 90%, 95%, 98%, 99%)
- Parameter of interest
- Calculated interval
- Order of subtraction when comparing two groups

Now we will estimate the true correlation between salinity and temperature of the Pacific Ocean.

```
set.seed(216)
regression_bootstrap_CI(T_degC~Salnty, # response ~ explanatory
  data = water, # Name of data set
  confidence_level = 0.95, # Confidence level as decimal
  summary_measure = "correlation", # Slope or correlation
  number_repetitions = 10000) # Number of simulated samples for bootstrap distribution
```



Confidence interval interpretation:

- How confident you are (e.g., 90%, 95%, 98%, 99%)
- Parameter of interest
- Calculated interval
- Order of subtraction when comparing two groups

1.2.3 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. Explain why theory-based methods should not be used to analyze the salinity study?
2. What is the proper notation for the population slope? Population correlation?

1.3 Activity 22: IPEDS (continued) — Linear Regression

1.3.1 Learning outcomes

- Identify and create appropriate summary statistics and plots given a data set with two quantitative variables.
- Use scatterplots to assess the relationship between two quantitative variables.
- Find the estimated line of regression using summary statistics and R linear model (`lm()`) output.
- Interpret the slope coefficient in context of the problem.

1.3.2 Terminology review

In today's activity, we will review summary measures and plots for two quantitative variables. Some terms covered in this activity are:

- Scatterplot
- Least-squares line of regression
- Slope and y -intercept
- Residuals

To review these concepts, see Chapter 6 & 7 in the textbook.

1.3.3 The Integrated Postsecondary Education Data System (IPEDS)

We will continue to assess the IPEDS data set collected on a subset of institutions that met the following selection criteria (Education Statistics 2018):

- Degree granting
- United States only
- Title IV participating
- Not for profit
- 2-year or 4-year or above
- Has full-time first-time undergraduates

Some of the variables collected and their descriptions are below. Note that several variables have missing values for some institutions (denoted by “NA”).

Variable	Description
UnitID	Unique institution identifier
Name	Institution name
State	State abbreviation
Sector	whether public or private
LandGrant	Is this a land-grant institution (Yes/No)
Size	Institution size category based on total student enrolled for credit, Fall 2018: Under 1,000, 1,000 - 4,999, 5,000 - 9,999, 10,000 - 19,999, 20,000 and above
Cost_OutofState	Cost of attendance for full-time out-of-state undergraduate students
Cost_InState	Cost of attendance for full-time in-state undergraduate students
Retention	Retention rate is the percent of the undergraduate students that re-enroll in the next year

Variable	Description
Graduation_Rate	6-year graduation rate for undergraduate students
SATMath_75	75th percentile Math SAT score
ACT_75	75th percentile ACT score

The code below reads in the needed data set, IPEDS_2018.csv, and filters out the 2-year institutions.

```
IPEDS <- read.csv("https://www.math.montana.edu/courses/s216/data/IPEDS_2018.csv")
IPEDS <- IPEDS %>%
  filter(Sector != "Public 2-year") #Filters the data set to remove Public 2-year
IPEDS <- IPEDS %>%
  filter(Sector != "Private 2-year") #Filters the data set to remove Private 2-year
IPEDS <- na.omit(IPEDS)
```

Notes on two quantitative variables

R Instructions

For this study, we will look to see if 75th percentile math SAT score can be used to predict the retention rate of 4-year higher education institutions.

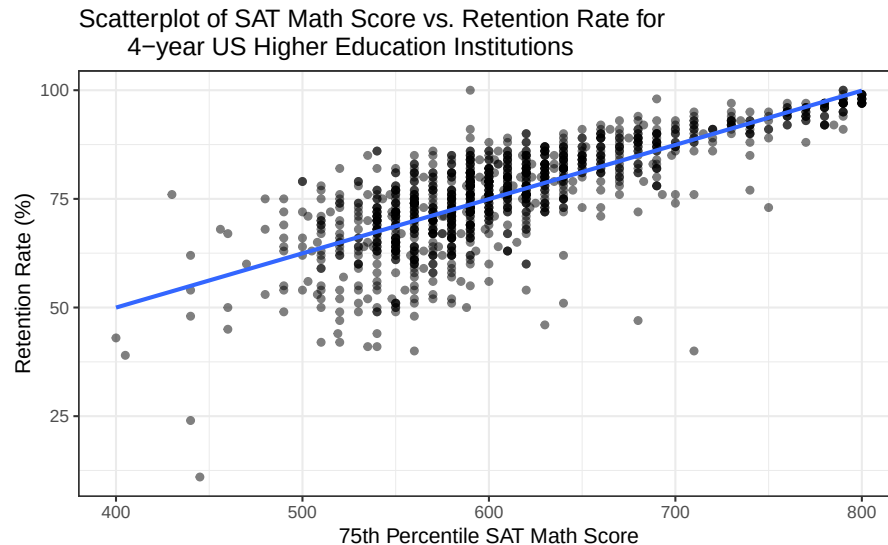
- Observational units:
- Explanatory variable:
 - Units?:
- Response variable:
 - Units?:
- Use the provided R script file to create a scatterplot to examine the relationship between 75th percentile Math SAT score by retention rate for 4-year US Higher Education Institutions by filling in the variable names for explanatory and response in line 16. Note, we are using the 75th percentile Math SAT score to predict the retention rate.
- Highlight and run lines 1–21.

```
IPEDS %>% # Data set pipes into...
  ggplot(aes(x = SATMath_75, y = Retention))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "75th Percentile SAT Math Score", # Label x-axis
```

```

y = "Retention Rate (%)", # Label y-axis
title = "Scatterplot of SAT Math Score vs. Retention Rate for
4-year US Higher Education Institutions" +
# Be sure to title your plots with the type of plot, observational units, variable(s)
geom_smooth(method = "lm", se = FALSE) + # Add regression line
theme_bw()

```



Assess the four features of the scatterplot that describe this relationship.

- Form (linear, non-linear)
- Direction (positive, negative)
- Strength
- Unusual observations or outliers

Slope

The linear model function in R (`lm()`) gives us the summary for the least squares regression line. The estimate for (**Intercept**) is the estimated y -intercept, b_0 for the line of least squares, and the estimate for **SATMath_75** (the x -variable name) is the value of b_1 , the estimated slope.

- Enter **Retention** for response and **SATMath_75** for explanatory in line 25
- Highlight and run lines 25–26 to fit the linear model.

```

# Fit linear model: y ~ x
IPEDSLM <- lm(Retention~SATMath_75, data=IPEDS)
round(summary(IPEDSLM)$coefficients, 5) # Display coefficient summary

```

```

#>           Estimate Std. Error  t value Pr(>|t|)
#> (Intercept)  0.05926    1.89844   0.03121   0.9751
#> SATMath_75   0.12481    0.00308  40.48525   0.0000

```

Write out the least squares regression line using the summary statistics provided above in context of the problem.

Interpret the value of slope in context of the problem.

Using the least squares line, predict the retention rate for a 4-year US higher education institution with a 75th percentile SAT Math score of 445.

Residuals

The model we are using assumes the relationship between the two variables follows a straight line. The residuals are the errors, or the variability in the response that hasn't been modeled by the regression line.

$$\Rightarrow \text{Residual} = \text{actual y value} - \text{predicted y value}$$

$$e = y - \hat{y}$$

Calculate the residual for a 4-year US higher education institution with a 75th percentile SAT Math score of 445 and a retention rate of 11%.

1. Circle the point (445, 11) on the scatterplot. Is the residual positive or negative?
2. Did the line of regression overestimate or underestimate the retention rate for this 4-year US higher education institution?

Correlation

The following output creates a correlation matrix between several pairs of quantitative variables.

```
IPEDS %>% # Data set pipes into
  select(c("Retention", "Cost_InState",
           "Graduation_Rate", "Salary",
           "SATMath_75", "ACT_75")) %>%
  cor(use="pairwise.complete.obs") %>%
  round(3)
```

```
#>           Retention Cost_InState Graduation_Rate Salary SATMath_75 ACT_75
#> Retention           1.000         0.388           0.832  0.698         0.767  0.768
#> Cost_InState        0.388         1.000           0.563  0.365         0.502  0.514
#> Graduation_Rate     0.832         0.563           1.000  0.683         0.817  0.833
#> Salary              0.698         0.365           0.683  1.000         0.747  0.706
#> SATMath_75          0.767         0.502           0.817  0.747         1.000  0.920
#> ACT_75              0.768         0.514           0.833  0.706         0.920  1.000
```

Report the value of correlation between the 75th SAT Math score and the retention rate.

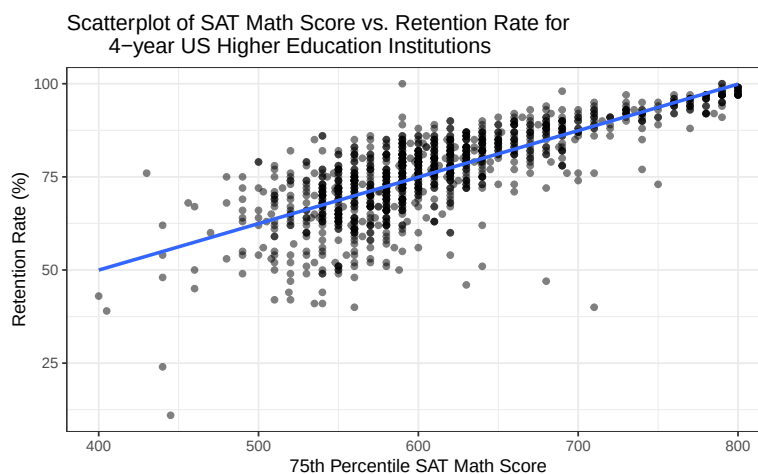
3. Which pair of variables have the strongest correlation?

4. Which pair of variables have the weakest correlation?

Leverage and influential observations

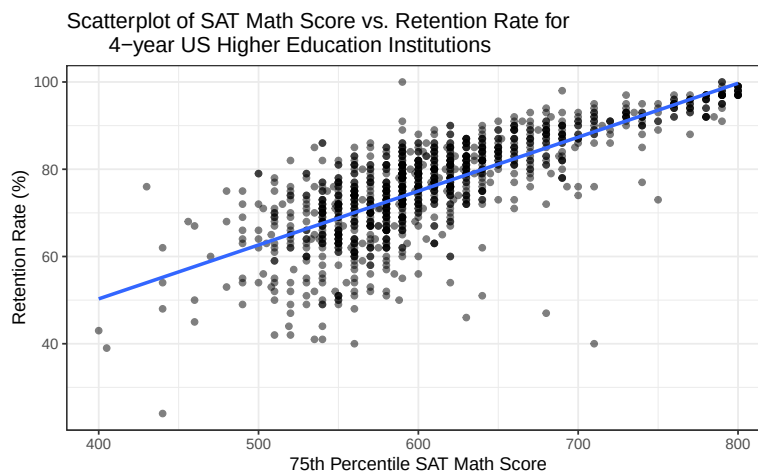
- Influential points are outliers that change the regression line; these points are far from the line of regression
- High leverage points are outliers that are extreme in the x-axis; these points are far from the mean of the variable on the x-axis

Let's review the scatterplot between the two variables.



5. Circle any values that may have high leverage. Put a square around any values that may be influential.

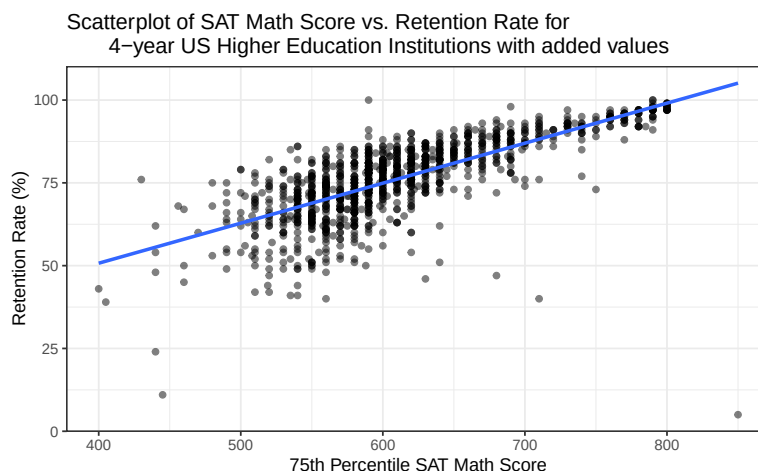
Earlier in the activity we identified a couple of observations that are potential outliers. One of these values is (445, 11). The following plot was created without this point included.



```
#>           Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   0.861      1.875   0.459   0.646
#> SATMath_75    0.124      0.003  40.598   0.000
```

6. Note that the slope of the line has not changed much between the original data set and the data set with the values removed. Would you consider this point to be influential?

7. In the next plot, note that one extreme values have been added to the data set. Circle this point on the plot.



8. The linear model and correlation output for the data set with the added point is provided below. Did the slope and/or value of correlation change in a noticeable way?

Linear model output for the data set with the added points:

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)      2.408      2.031   1.186   0.236
#> IPEDS.SATMath_75  0.121      0.003  36.650   0.000
```

Correlation between the 75th percentile Math SAT score and retention rate for the data set with the added point:

```
#> [1] 0.7347292
```

9. Earlier in the semester, we learned that the median and the IQR are resistant to outliers. In other words, they are not influenced by the presence of outliers. Are the values of slope and correlation resistant to outliers? Explain your answer using the previous examples in this activity.

1.3.4 Take-home messages

- Two quantitative variables are graphically displayed in a scatterplot. The explanatory variable is on the x -axis and the response variable is on the y -axis. When describing the relationship between two quantitative variables we look at the form (linear or non-linear), direction (positive or negative), strength, and for the presence of outliers.
- There are three summary statistics used to summarize the relationship between two quantitative variables: correlation (r), slope of the regression line (b_1), and the coefficient of determination (r^2). In this activity, we reviewed the first two summary measures, correlation and slope. We will explore the coefficient of determination in the next activity.

3. Points that are far from the mean of the predictor (x-axis variable) and that do not follow the pattern of the relationship can impact the line and be influential.

1.3.5 Additional notes

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

1.4 Activity 20: Penguins

1.4.1 Learning outcomes

- Identify and create appropriate summary statistics and plots given a data set with two quantitative variables.
- Find the estimated line of regression using summary statistics and R linear model (`lm()`) output.
- Interpret the slope coefficient in context of the problem.
- Calculate and interpret r^2 , the coefficient of determination, in context of the problem.
- Find the correlation coefficient from R output or from r^2 and the sign of the slope.

1.4.2 Terminology review

In today's activity, we will review summary measures and plots for two quantitative variables. Some terms covered in this activity are:

- Least-squares line of regression
- Slope and y -intercept
- Residuals
- Correlation (r)
- Coefficient of determination (r -squared)

To review these concepts, see Chapter 6 in the textbook.

1.4.3 Penguins

The Palmer Station Long Term Ecological Research Program sampled three penguin species on islands in the Palmer Archipelago in Antarctica over a period of three summers. Researchers took various body measurements on the penguins, including flipper length (mm), bill depth (mm), bill length (mm), sex, species (Adelie, Chinstrap, Gentoo), and body mass (g). The researchers were interested in the relationship between bill depth and body mass and wondered if flipper length could be used to accurately predict the body mass of these three penguin species.

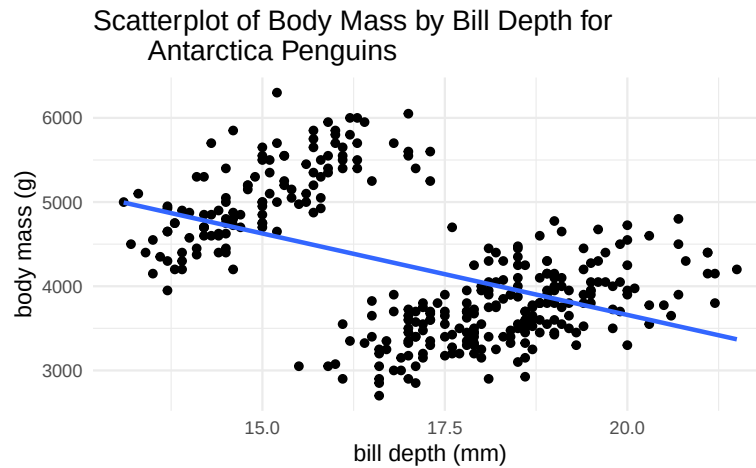
- Upload and import the `Antarctica_Penguins` csv file and the provided R script file for this activity. Enter the name of the data set for `datasetname` in the R script file in line 6

First we will create a scatterplot of the bill depth and body mass. Notice that we are using bill depth to predict body mass. This makes bill depth the explanatory variable.

- Enter `bill_depth` for explanatory and `body_mass` for response in line 9
- Highlight and run lines 1–15 in the R script file.

```
penguins <- read.csv("data/Antarctica_Penguins.csv") #Creates the object penguins
penguins <- na.omit(penguins)
penguins %>%
  ggplot(aes(x = bill_depth, y = body_mass))+ # Specify variables
  geom_point() + # Add scatterplot of points
  labs(x = "bill depth (mm)", # Label x-axis
       y = "body mass (g)", # Label y-axis
       title = "Scatterplot of Body Mass by Bill Depth for
               Antarctica Penguins") +
```

```
# Be sure to title your plots
geom_smooth(method = "lm", se = FALSE) # Add regression line
```



1. Describe the relationship, using the four characteristics of scatterplots, between bill depth and body mass.

Slope of the Least Squares Linear Regression Line

There are three summary measures calculated from two quantitative variables: slope, correlation, and the coefficient of determination. We will again start with an assessment of the regression slope.

- Enter `body_mass` for response and `bill_depth` for explanatory in line ...
- Highlight and run lines ... to fit the linear model.

```
# Fit linear model: y ~ x
PenguinsLM <- lm(body_mass~bill_depth, data=penguins)
round(summary(PenguinsLM)$coefficients,5) # Display coefficient summary
```

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  7528.6205   341.58707  22.04012      0
#> bill_depth   -193.4424    19.77623  -9.78156      0
```

2. Write out the least squares regression line using the summary statistics from the R output in context of the problem.
3. Interpret the value of slope in context of the problem.

4. Predict the body mass of a penguin with a bill depth of 17.3 mm.
5. Calculate the residual for a penguin with a bill depth of 17.3 mm and a body mass of 5600 g.

Correlation

Additionally, we can find the value of correlation.

The following output creates a correlation matrix between several pairs of quantitative variables.

```
penguins %>% # Data set pipes into
  select(c("bill_length", "bill_depth",
           "flipper_length", "body_mass")) %>%
  cor(use="pairwise.complete.obs") %>%
  round(3)
```

```
#>           bill_length bill_depth flipper_length body_mass
#> bill_length           1.000    -0.229           0.652     0.589
#> bill_depth          -0.229     1.000          -0.579    -0.473
#> flipper_length       0.652    -0.579           1.000     0.873
#> body_mass            0.589    -0.473           0.873     1.000
```

6. What is the value of correlation between bill depth and body mass?

Coefficient of determination (squared correlation)

Another summary measure used to explain the linear relationship between two quantitative variables is the coefficient of determination (r^2). The coefficient of determination, r^2 , can also be used to describe the strength of the linear relationship between two quantitative variables. The value of r^2 (a value between 0 and 1) represents the **proportion of variation in the response that is explained by the least squares line with the explanatory variable**. There are two ways to calculate the coefficient of determination:

Square the correlation coefficient: $r^2 = (r)^2$

Use the variances of the response and the residuals: $r^2 = \frac{s_y^2 - s_{RES}^2}{s_y^2} = \frac{SST - SSE}{SST}$

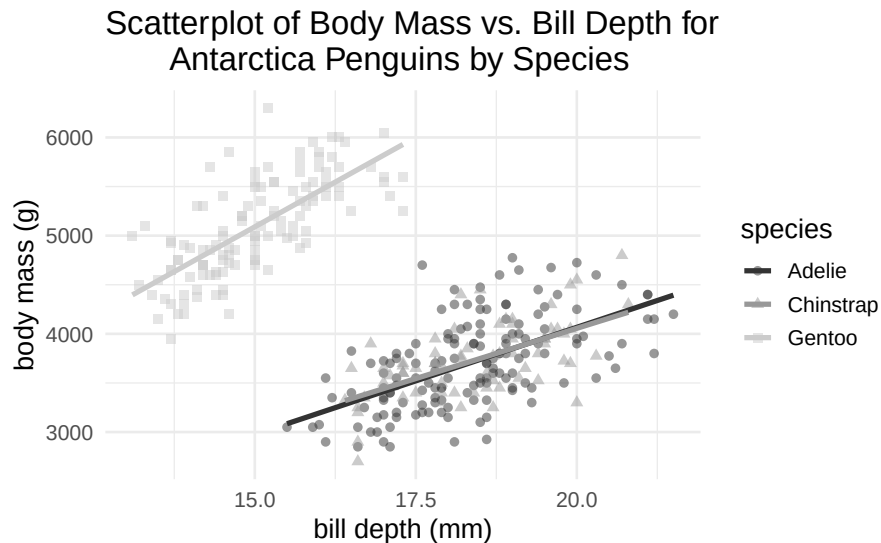
Later in this activity, we will explore more about what the coefficient of determination measures.

7. Use the correlation, r , found in the correlation matrix, to calculate the coefficient of determination between bill depth and body mass, r^2 , for Antarctica penguins.
8. The coefficient of determination between bill depth and flipper length is 33.5%. Calculate the correlation between these variables from this value. How can we tell if the value of correlation should be positive or negative?

Multivariable plots

Recall earlier we noted two clouds of data in the scatterplot. Let's explore another variable that may account for the two clouds of point shown on the scatterplot. When adding a categorical predictor, we can add that variable as shape and/or color to the plot.

```
penguins %>% # Data set pipes into...
  ggplot(aes(x = bill_depth, y = body_mass, shape = species, color=species))+
  # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "bill depth (mm)", # Label x-axis
       y = "body mass (g)", # Label y-axis
       title = "Scatterplot of Body Mass vs. Bill Depth for
               Antarctica Penguins by Species") + # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) + # Add regression line
  scale_color_grey()
```



Note that the relationship between bill depth and body mass is now positive when considering the relationship within each species. This is an example of what?

Does the relationship between bill depth and body mass for Antarctica penguins change depending on the species?

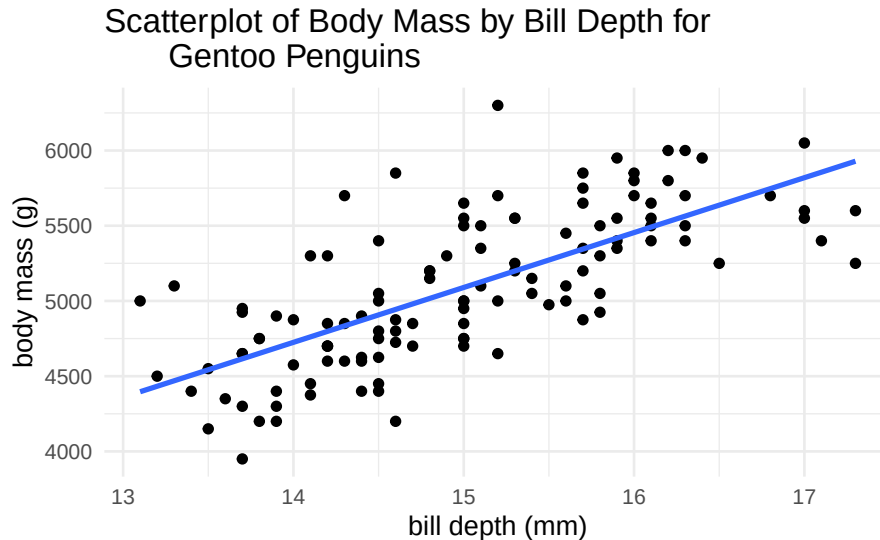
For the remainder of this study, we will only look at the Gentoo species.

Coefficient of determination (revisited)

We will now revisit the coefficient of determination for the relationship between bill depth and body mass for just the Gentoo species. As a reminder, these are the formulas used to calculate the coefficient of determination:

Square the correlation coefficient: $r^2 = (r)^2$

Use the variances of the response and the residuals: $r^2 = \frac{s_y^2 - s_{RES}^2}{s_y^2} = \frac{SST - SSE}{SST}$



```
penguin_sub %>% # Data set pipes into
  select(c("bill_length", "bill_depth",
           "flipper_length", "body_mass")) %>%
  cor(use="pairwise.complete.obs") %>%
  round(5)
```

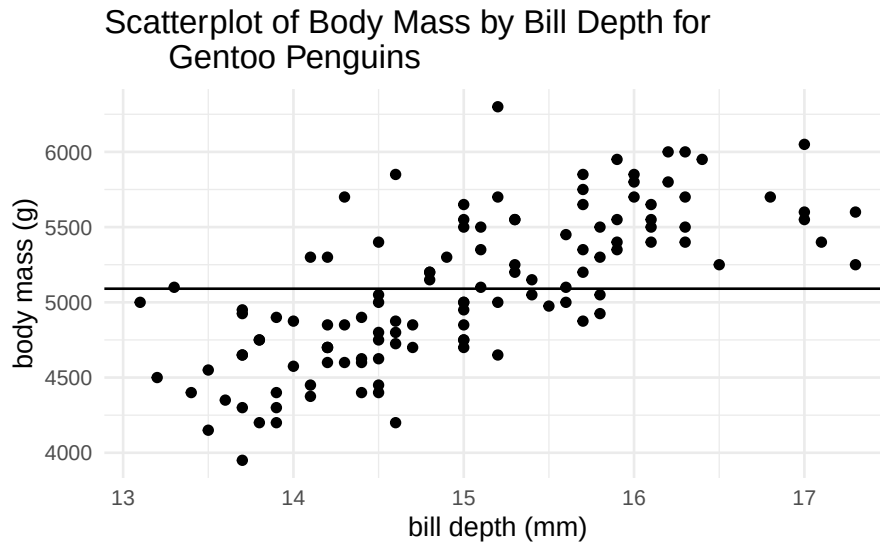
```
#>          bill_length bill_depth flipper_length body_mass
#> bill_length      1.00000    0.64409    0.66179    0.66707
#> bill_depth      0.64409    1.00000    0.70891    0.71827
#> flipper_length  0.66179    0.70891    1.00000    0.71087
#> body_mass      0.66707    0.71827    0.71087    1.00000
```

9. Use the correlation, r , found in the correlation matrix, to calculate the coefficient of determination between bill depth and body mass, r^2 , for the Gentoo species.

The variance of the response variable, body mass (g), is $s_{\text{body mass}}^2 = 249759.1 \text{ g}^2$, the variance of the explanatory variable, bill depth (mm), is $s_{\text{bill depth}}^2 = 0.968145 \text{ mm}^2$, and the variance in the residuals is $s_{RES}^2 = 120907.1 \text{ g}^2$. Use these values to calculate the coefficient of determination.

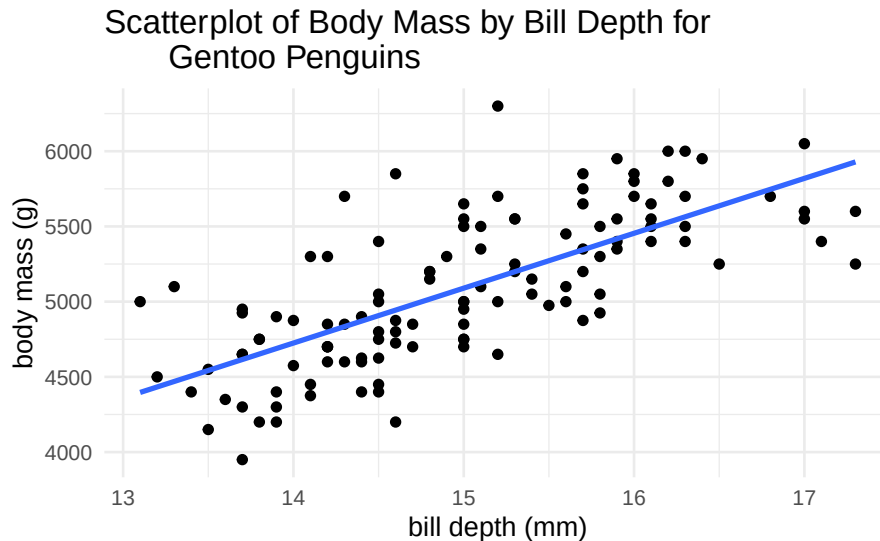
In the next part of the activity we will explore what the coefficient of determination measures.

In the first scatterplot, we see the data plotted with a horizontal line. Note that the regression line in this plot has a slope of zero; this assumes there is no relationship between bill depth and body mass. The value of the y-intercept, 5090.625, is the mean of the response variable when there is no relationship between the two variables. To find the sum of squares total (SST) we find the residual ($residual = y - \hat{y}$) for each response value from the horizontal line (from the value of 5090.625). Each residual is squared and the sum of the squared values is calculated. The SST gives the **total variability in the response variable, body mass**.



The calculated value for the SST is 29721328.125.

This next scatterplot, shows the plotted data with the best fit regression line. This is the line of best fit between bill depth and body mass and has the smallest sum of squares error (SSE). The SSE is calculated by finding the residual from each response value to the regression line. Each residual is squared and the sum of the squared values is calculated.



The calculated value for the SSE is 14387948.480.

Calculate the value for r^2 using the values for SST and SSE provided below each of the previous graphs.

Interpretation of the coefficient of determination in context of the problem.

1.4.4 Take-home messages

1. The sign of correlation and the sign of the slope will always be the same. The closer the value of correlation is to -1 or $+1$, the stronger the linear relationship between the explanatory and the response variable.
2. The coefficient of determination multiplied by 100 ($r^2 \times 100$) measures the percent of variation in the response variable that is explained by the relationship with the explanatory variable. The closer the value of the coefficient of determination is to 100%, the stronger the relationship.
3. We can use the line of regression to predict values of the response variable for values of the explanatory variable. Do not use values of the explanatory variable that are outside of the range of values in the data set to predict values of the response variable (reflect on why this is true.). This is called **extrapolation**.

1.4.5 Additional notes

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

1.5 Activity 24: Golf Driving Distance

1.5.1 Learning outcomes

- Given a research question involving two quantitative variables, construct the null and alternative hypotheses in words and using appropriate statistical symbols.
- Assess the conditions to use the normal distribution model for a slope.
- Find the T test statistic (T-score) for a slope based off of `lm()` output in R.
- Find, interpret, and evaluate the p-value for a theory-based hypothesis test for a slope.
- Create and interpret a theory-based confidence interval for a slope.
- Use a confidence interval to determine the conclusion of a hypothesis test.

1.5.2 Terminology review

In this week's in-class activity, we will use theory-based methods for hypothesis tests and confidence intervals for a linear regression slope. Some terms covered in this activity are:

- Slope
- Regression line

To review these concepts, see Chapter 21 in the textbook.

1.5.3 Golf driving distance

In golf the goal is to complete a hole with as few strokes as possible. A long driving distance to start a hole can help minimize the strokes necessary to complete the hole, as long as that drive stays on the fairway. Data were collected on 354 PGA and LPGA players in 2008 (*Average Driving Distance and Fairway Accuracy* 2008). For each player, the average driving distance (yards), fairway accuracy (percentage), and league (PGA or LPGA) was measured. Use these data to assess, "Does a professional golfer give up accuracy when they hit the ball farther?"

- Observational units:
- Explanatory variable:
 - Units?:
- Response variable:
 - Units?:

R Instructions

- Download the R script file from Canvas and open in the RStudio server

```
# Read in data set
golf <- read.csv("https://math.montana.edu/courses/s216/data/golf.csv")
```

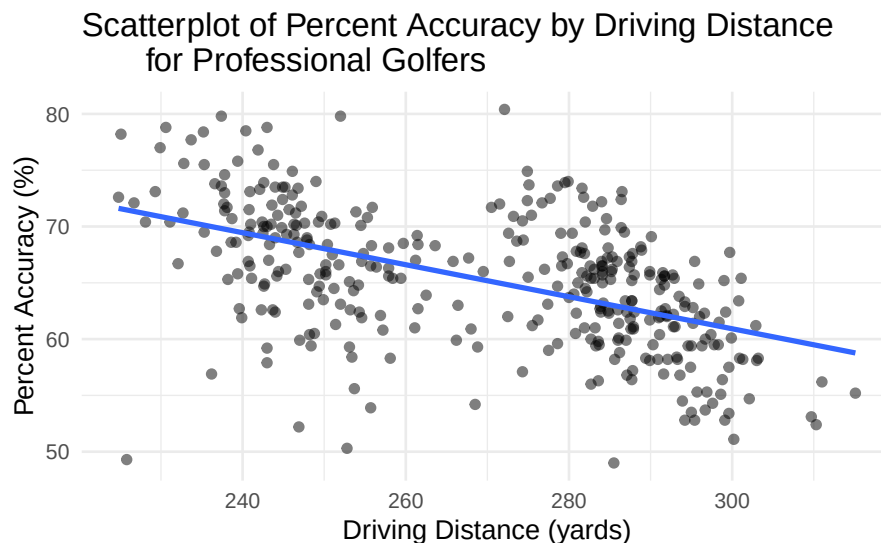
Plot review.

To create a scatterplot showing the relationship between the driving distance and percent accuracy for professional golfers:

- Enter the name of the explanatory and response in line 10

- Highlight and run lines 1 - 16

```
golf %>% # Pipe data set into...
ggplot(aes(x = Driving_Distance, y = Percent_Accuracy))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "Driving Distance (yards)", # Label x-axis
       y = "Percent Accuracy (%)", # Label y-axis
       title = "Scatterplot of Percent Accuracy by Driving Distance
for Professional Golfers") +
  # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) # Add regression line
```

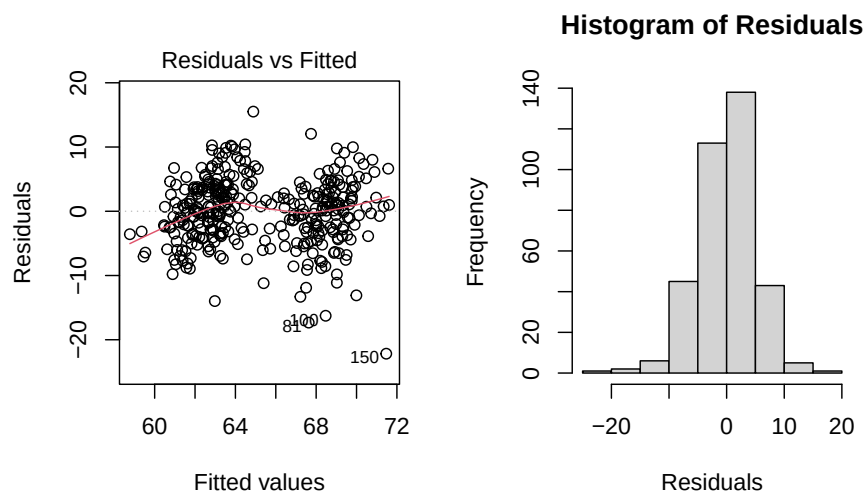


Conditions for the least squares line

When performing inference on a least squares line, the follow conditions are generally required:

- *Independent observations* (for both simulation-based and theory-based methods): individual data points must be independent.
 - Check this assumption by investigating the sampling method and determining if the observational units are related in any way.
- *Linearity* (for both simulation-based and theory-based methods): the data should follow a linear trend.
 - Check this assumption by examining the scatterplot of the two variables, and a scatterplot of the residuals (on the y -axis) versus the fitted values (on the x -axis). The pattern in the residual plot should display a horizontal line.
- *Constant variability* (for theory-based methods only): the variability of points around the least squares line remains roughly constant
 - Check this assumption by examining a scatterplot of the residuals (on the y -axis) versus the fitted values (on the x -axis). The variability in the residuals around zero should be approximately the same for all fitted values.
- *Nearly normal residuals* (for theory-based methods only): residuals must be nearly normal.
 - Check this assumption by examining a histogram of the residuals, which should appear approximately normal.

The scatterplot generated earlier and the residual plots shown below will be used to assess these conditions for approximating the data with the t -distribution.



Verify the conditions are met to use theory-based methods

Ask a research question

Use these data to assess, “Does a professional golfer give up accuracy when they hit the ball farther?”

Parameter of interest in context of the study:

Null Hypothesis (in words):

Null Hypothesis (in notation):

Alternative Hypothesis (in words):

Alternative Hypothesis (in notation):

Summarize and visualize the data

The linear model output for this study is shown below.

```
lm.golf <- lm(Percent_Accuracy~Driving_Distance, data=golf) # lm(response~explanatory)
round(summary(lm.golf)$coefficients, 3)
```

```
#>               Estimate Std. Error t value Pr(>|t|)
#> (Intercept)    103.586      3.329   31.119      0
#> Driving_Distance  -0.142      0.012  -11.553      0
```

1. Report and interpret the summary statistic (sample slope) for the linear relationship between driving distance and percent accuracy of golfers. Use proper notation.

Use statistical inferential methods to draw inferences from the data

Hypothesis test To find the value of the standardized statistic to test the slope we will use,

$$T = \frac{\text{slope estimate} - \text{nullvalue}}{SE} = \frac{b_1 - 0}{SE(b_1)}.$$

We will use the linear model R output above to get the estimate for slope and the standard error of the slope.

Calculate the standardized statistic for slope.

Report the p-value to answer the research question.

Confidence interval Recall that a confidence interval is calculated by adding and subtracting the margin of error to the point estimate.

$$\text{point estimate} \pm t^* \times SE(\text{estimate}).$$

When the point estimate is a regression slope, this formula becomes

$$b_1 \pm t^* \times SE(b_1).$$

The t^* multiplier comes from a t -distribution with $n - 2$ degrees of freedom. The sample size for this study is 354 so we will use the degrees of freedom 352 ($n - 2$).

- Enter the percentile needed to find the multiplier for a 95% confidence interval for xx
- Enter the degrees of freedom for yy
- Highlight and run line 34

```
qt(xx, yy, lower.tail = TRUE) # 95% t* multiplier
```

Calculate the 95% confidence interval for the true slope.

Interpret the 95% confidence interval in context of the problem.

Communicate the results and answer the research question

2. Write a conclusion to answer the research question in context of the problem.

Simulation-based hypothesis test

Let's start by thinking about how one simulation would be created on the null distribution using cards. First, we would write the values for the response variable, total length, on each card. Next, we would shuffle these y values while keeping the x values (explanatory variable) in the same order. Then, find the line of regression for the shuffled (x, y) pairs and calculate either the slope or correlation of the shuffled sample.

We will use the `regression_test()` function in R (in the `catstats` package) to simulate the null distribution of shuffled slopes (or shuffled correlations) and compute a p-value. We will need to enter the response variable name and the explanatory variable name for the formula, the data set name, the summary measure for the test (either slope or correlation), number of repetitions, the sample statistic (value of slope or correlation), and the direction of the alternative hypothesis.

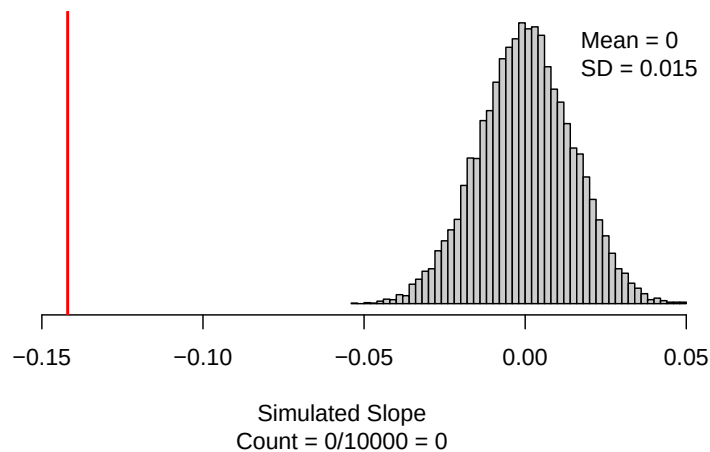
3. What inputs should be entered for each of the following to create the simulation to test regression slope?
 - Direction ("greater", "less", or "two-sided"):

- Summary measure (choose "slope" or "correlation"):
- As extreme as (enter the value for the sample slope):
- Number of repetitions:

Using the R script file for this activity...

- Enter your answers for question 3 in place of the xx's to produce the null distribution with 10000 simulations.
- Highlight and run lines 38–43.

```
regression_test(Percent_Accuracy~Driving_Distance, # response ~ explanatory
  data = golf, # Name of data set
  direction = "less", # Sign in alternative ("greater", "less", "two-sided")
  summary_measure = "slope", # "slope" or "correlation"
  as_extreme_as = -0.142, # Observed slope or correlation
  number_repetitions = 10000) # Number of simulated samples for null distribution
```

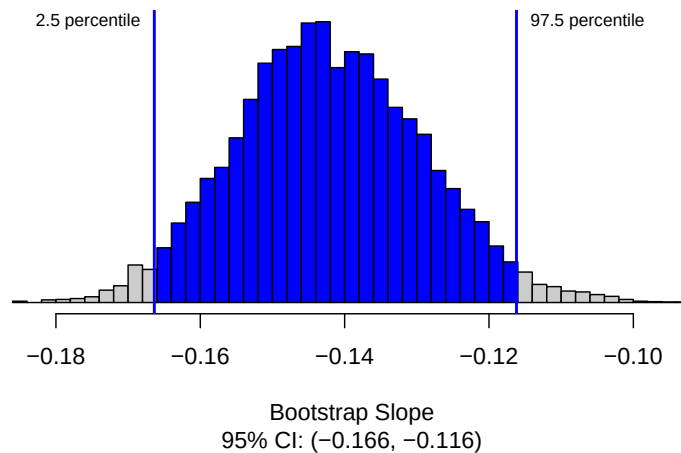


Simulation-based confidence interval

We will use the `regression_bootstrap_CI()` function in R (in the `catstats` package) to simulate the bootstrap distribution of sample slopes (or sample correlations) and calculate a confidence interval.

- Fill in the missing values in the provided R script file to find a 95% confidence interval for slope.
- Highlight and run lines 42–46.

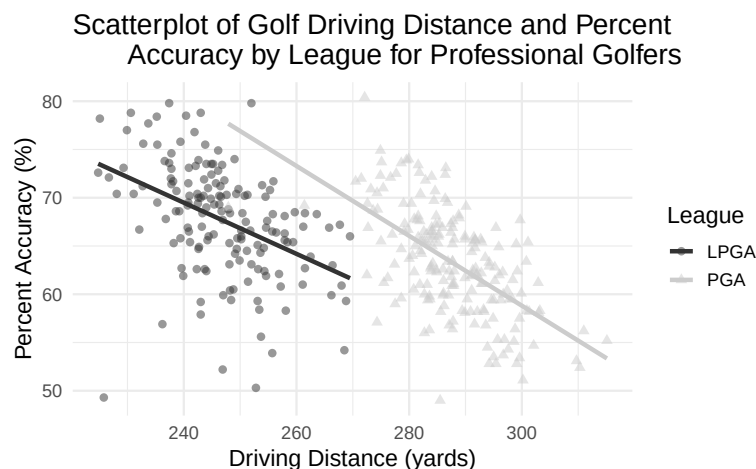
```
regression_bootstrap_CI(Percent_Accuracy~Driving_Distance, # response ~ explanatory
  data = golf, # Name of data set
  confidence_level = 0.95, # Confidence level as decimal
  summary_measure = "slope", # Slope or correlation
  number_repetitions = 10000) # Number of simulated samples for bootstrap distribution
```



Multivariable plots

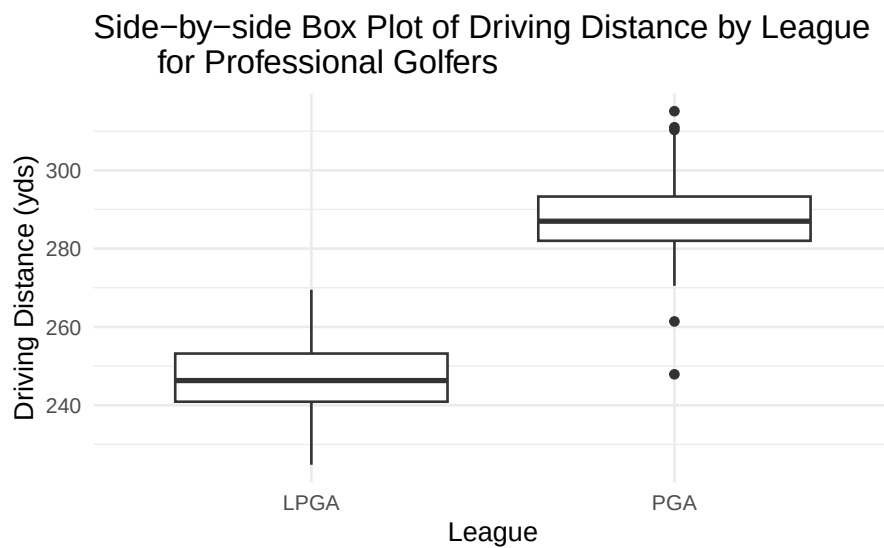
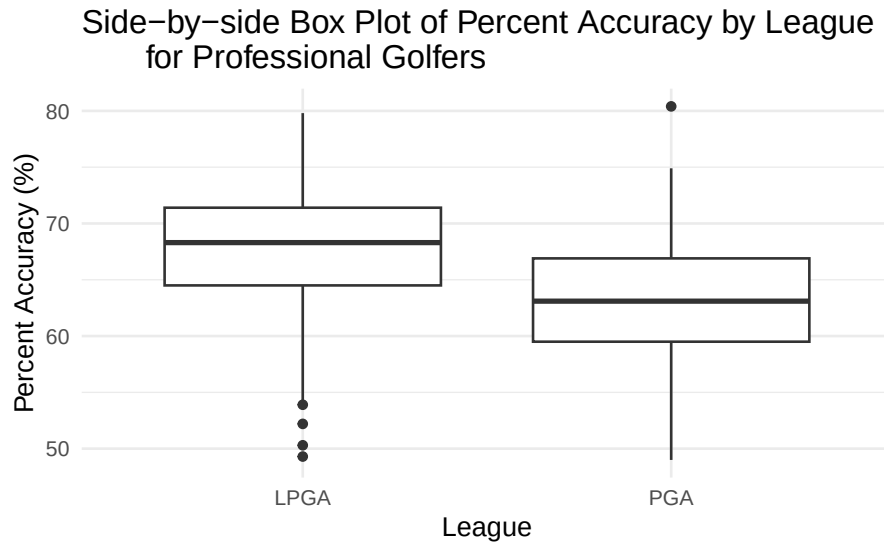
Another variable that may affect the percent accuracy is the which league the golfer is part of. We will look at how this variable may change the relationship between driving distance and percent accuracy.

```
golf %>%
  ggplot(aes(x = Driving_Distance, y = Percent_Accuracy, color=League))+ # Specify variables
  geom_point(aes(shape = League), size = 2, alpha=0.5) + # Add scatterplot of points
  labs(x = "Driving Distance (yards)", # Label x-axis
       y = "Percent Accuracy (%)", # Label y-axis
       color = "League", shape = "League",
       title = "Scatterplot of Golf Driving Distance and Percent
               Accuracy by League for Professional Golfers") + # Be sure to title your plots
  geom_smooth(method = "lm", se = FALSE) + # Add regression line
  scale_color_grey()
```



- Does the association between driving distance and percent accuracy change depending on which league the golfer is a part of? Explain your answer.

5. Explain the association between league and each of the other two variables. Use the following plots in addition to the scatterplot from Q4 to explain your answer.



1.5.4 Take-home messages

1. To check the validity conditions for using theory-based methods we must use the residual diagnostic plots to check for normality of residuals and constant variability, and the scatterplot to check for linearity.
2. To interpret a confidence interval for the slope, think about how to interpret the sample slope and use that information in the confidence interval interpretation for slope.
3. Use the explanatory variable row in the linear model R output to obtain the slope estimate (**estimate** column) and standard error of the slope (**Std. Error** column) to calculate the standardized slope, or T-score. The calculated T-score should match the **t value** column in the explanatory variable row. The standardized slope tells the number of standard errors the observed slope is above or below 0.
4. The explanatory variable row in the linear model R output provides a **two-sided** p-value under the **Pr(>|t|)** column.
5. The standardized slope is compared to a t -distribution with $n - 2$ degrees of freedom in order to obtain a p-value. The t -distribution with $n - 2$ degrees of freedom is also used to find the appropriate multiplier for a given confidence level.

1.5.5 Additional notes

Use this space to summarize your thoughts and take additional notes on this week's activity and material covered.

1.6 Module 13 Lab: Big Mac Index

1.6.1 Learning outcomes

- Given a research question involving two quantitative variables, construct the null and alternative hypotheses in words and using appropriate statistical symbols.
- Assess the conditions to determine in theory or simulation-based methods should be used.
- Find, interpret, and evaluate the p-value for a hypothesis test for a slope or correlation.
- Create and interpret a confidence interval for a slope or correlation.

1.6.2 Big Mac Index

Can the relative cost of a Big Mac across different countries be used to predict the Gross Domestic Product (GDP) per person for that country? The GDP per person (\$) the adjusted dollar equivalent to purchase a Big Mac (\$) was found on a random sample of 55 countries in January of 2022. The cost of a Big Mac in each country was adjusted to US dollars based on current exchange rates. Is there evidence of a positive relationship between Big Mac cost (`dollar_price`) and the GDP per person (`GDP_dollar`)?

- Upload and open the R script file for Module 13 lab.
- Upload the csv file, `big_mac_adjusted_index_22.csv`.
- Enter the name of the data set for `datasetname` in the R script file in line 9.
- Highlight and run lines 1–9 to load the data.

```
# Read in data set
mac <- read.csv("datasetname.csv")
```

Conditions for the least squares line

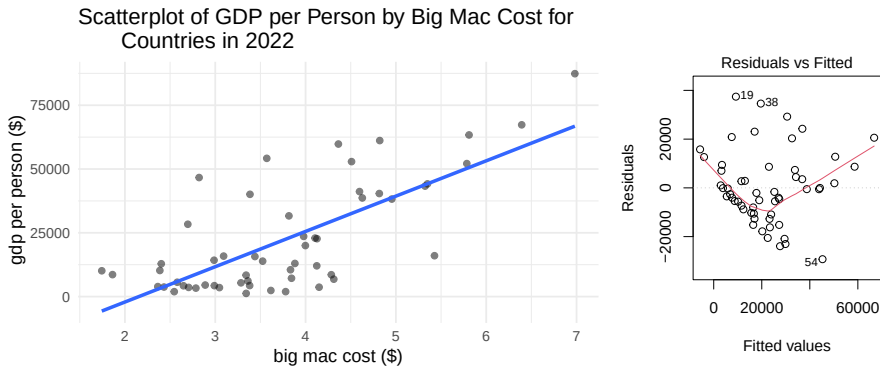
1. Is there independence between the responses for the observational units? Justify your answer.

The following code will create a scatterplot and residuals vs. fitted plot to check for linearity.

- Highlight and run lines 14–24

```
par(mfrow=c(1,2)) # Set graphics parameters to plot 2 plots in 1 row
#Scatterplot
mac %>% # Pipe data set into...
  ggplot(aes(x = dollar_price, y = GDP_dollar))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "big mac cost ($)", # Label x-axis
       y = "gdp per person ($)", # Label y-axis
       title = "Scatterplot of GDP per Person by Big Mac Cost for
               Countries in 2022") +
  geom_smooth(method = "lm", se = FALSE) # Add regression line

#Diagnostic plots to check linearity
macLM <- lm(GDP_dollar~dollar_price, data = mac) # Fit linear regression model
plot(macLM, which=1) # Residual vs fitted values
```



2. Is the linearity condition met to use regression methods to analyze the data? Justify your answer.

One way to “fix” the issues with linearity is to take the log of either the response, explanatory or both variables. Let’s check for linearity after log transforming the response variable.

- Enter the `dollar_price` for explanatory and `log_GDP` for response in line 32
- Enter a title and x and y axes labels
- Highlight and run lines 31–40

```
#Scatterplot
mac %>% # Pipe data set into...
  ggplot(aes(x = dollar_price, y = log_GDP))+ # Specify variables
  geom_point(alpha=0.5) + # Add scatterplot of points
  labs(x = "include an x-axis label (don't forget units)", # Label x-axis
       y = "include an y-axis label (don't forget units)", # Label y-axis
       title = "A title should include type of plot, OUs, variables") +
  geom_smooth(method = "lm", se = FALSE) # Add regression line
#Diagnostic plots to check linearity
macLM <- lm(log_GDP~dollar_price, data = mac) # Fit linear regression model
plot(macLM, which=1) # Residual vs fitted values
```

3. Assess linearity using the scatterplot and the residuals vs. fitted plot for the log transformed data.

For the remainder of this lab we will use the log transformed response variable, `log_GDP`.

- Highlight and run lines 44–427 to produce the histogram of residuals needed to assess normality of residuals condition

```
#Diagnostic plots
bigmacLM <- lm(log_GDP~dollar_price, data = mac) # Fit linear regression model
hist(bigmacLM$resid, xlab="Residuals", ylab="Frequency",
     main = "Histogram of Residuals") # Histogram of residuals
```

4. Are the conditions met to use the *t*-distribution to approximate the sampling distribution of the standardized statistic? Justify your answer.

Ask a research question

5. Write out the null and alternative hypotheses in notation to test *slope* between Big Mac cost and log GDP.

$H_0 :$

$H_A :$

Summary statistics

- To find the correlation between the variables, `log_GDP` and `dollar_price` enter the name of the explanatory and response variable
- Highlight and run lines 51–54 in the R script file.

```
mac %>%  
  select(c("explanatory", "response")) %>%  
  cor(use="pairwise.complete.obs") %>%  
  round(5)
```

6. Report the value of correlation between the variables.
7. Calculate the value of the coefficient of determination between `log_GDP` and `dollar_price`.
8. Interpret the value of the coefficient of determination in context of the problem.

In the next part of the activity we will assess the linear model between Big Mac cost and log GDP.

- Enter the variable `log_GDP` for `response` and the variable `dollar_price` for `explanatory` in line 22.
- Highlight and run lines 22–23 to get the linear model output.

```
# Fit linear model: y ~ x  
bigmacLM <- lm(response~explanatory, data=mac)  
round(summary(bigmacLM)$coefficients,3) # Display coefficient summary
```

9. Give the value of the slope of the regression line. Interpret this value in context of the problem.

Use statistical inferential methods to draw inferences from the data

Hypothesis test

Use the `regression_test()` function in R (in the `catstats` package) to simulate the null distribution of sample **slopes** and compute a p-value. We will need to enter the response variable name and the explanatory variable name for the formula, the data set name (identified above as `mac`), the summary measure used for the test, number of repetitions, the sample statistic (value of correlation), and the direction of the alternative hypothesis.

The response variable name is `log_GDP` and the explanatory variable name is `dollar_price`.

Using the R script file for this activity, produce the null distribution with 10000 simulations.

- Highlight and run lines 59–64.

```
regression_test(response~explanatory, # response ~ explanatory
  data = mac, # Name of data set
  direction = "xx", # Sign in alternative ("greater", "less", "two-sided")
  summary_measure = "xx", # "slope" or "correlation"
  as_extreme_as = xx, # Observed slope or correlation
  number_repetitions = 10000) # Number of simulated samples for null distribution
```

10. Report the p-value from the R output.

Simulation-based confidence interval

We will use the `regression_bootstrap_CI()` function in R (in the `catstats` package) to simulate the bootstrap distribution of sample **slopes** and calculate a 90% confidence interval.

- Fill in the `xx`'s in the the provided R script file to find a 90% confidence interval.
- Highlight and run lines 69–73.

```
regression_bootstrap_CI(response~explanatory, # response ~ explanatory
  data = mac, # Name of data set
  confidence_level = xx, # Confidence level as decimal
  summary_measure = "xx", # Slope or correlation
  number_repetitions = 10000) # Number of simulated samples for bootstrap distribution
```

11. Report the bootstrap 90% confidence interval in interval notation.

12. Write a paragraph summarizing the results of the study as if you are reporting these results in your local newspaper. **Upload a copy of your paragraph to Gradescope for your group.** Be sure to describe:

- Introduction statement
 - Research question
 - Observational units
 - Variables (roll & type)
 - Sampling method
 - Study Design
- Summary statistic and interpretation
 - Summary measure (in context)
 - Value of the statistic
- P-value and interpretation
 - Statement about probability or proportion of samples
 - Statistic (summary measure and value)

- Direction of the alternative
- Null hypothesis (in context)
- Confidence interval and interpretation
 - How confident you are (e.g., 90%, 95%, 98%, 99%)
 - Parameter of interest
 - Calculated interval
 - Order of subtraction when comparing two groups
- Conclusion (written to answer the research question)
 - Amount of evidence
 - Parameter of interest
 - Direction of the alternative hypothesis
- Scope of inference
 - To what group of observational units do the results apply (target population or observational units similar to the sample)?
 - What type of inference is appropriate (causal or non-causal)?

Paragraph:

- Average Driving Distance and Fairway Accuracy*. 2008. <https://www.pga.com/> and <https://www.lpga.com/>.
- Banton, et al, S. 2022. “Jog with Your Dog: Dog Owner Exercise Routines Predict Dog Exercise Routines and Perception of Ideal Body Weight.” *PLoS ONE* 17(8).
- Bhavsar, et al, A. 2022. “Increased Risk of Herpes Zoster in Adults ≥ 50 Years Old Diagnosed with COVID-19 in the United States.” *Open Forum Infectious Diseases* 9(5).
- Bulmer, M. n.d. *Islands in Schools Project*. <https://sites.google.com/site/islandsinschoolsprojectwebsite/home>.
- Bureau of Transportation Statistics*. 2019. <https://www.bts.gov/>.
- Child Health and Development Studies*. n.d. <https://www.chdstudies.org/>.
- Darley, J. M., and C. D. Batson. 1973. “”From Jerusalem to Jericho”: A Study of Situational and Dispositional Variables in Helping Behavior.” *Journal of Personality and Social Psychology* 27: 100–108.
- Davis, Smith, A. K. 2020. “A Poor Substitute for the Real Thing: Captive-Reared Monarch Butterflies Are Weaker, Paler and Have Less Elongated Wings Than Wild Migrants.” *Biology Letters* 16.
- Du Toit, et al, G. 2015. “Randomized Trial of Peanut Consumption in Infants at Risk for Peanut Allergy.” *New England Journal of Medicine* 372.
- Edmunds, et al, D. 2016. “Chronic Wasting Disease Drives Population Decline of White-Tailed Deer.” *PLoS ONE* 11(8).
- Education Statistics, National Center for. 2018. *IPEDS*. <https://nces.ed.gov/ipeds/>.
- Great Britain Married Couples: Great Britain Office of Population Census and Surveys*. n.d. <https://discover.y.nationalarchives.gov.uk/details/r/C13351>.
- Group, TODAY Study. 2012. “A Clinical Trial to Maintain Glycemic Control in Youth with Type 2 Diabetes.” *New England Journal of Medicine* 366: 2247–56.
- Hamblin, J. K., K. Wynn, and P. Bloom. 2007. “Social Evaluation by Preverbal Infants.” *Nature* 450 (6288): 557–59.
- Hirschfelder, A., and P. F. Molin. 2018. *I Is for Ignoble: Stereotyping Native Americans*. Retrieved from <https://www.ferris.edu/HTMLS/news/jimcrow/native/homepage.htm>.
- Hutchison, R. L., and M. A. Hirthler. 2013. “Upper Extremity Injuries in Homer’s Iliad.” *Journal of Hand Surgery (American Volume)* 38: 1790–93.
- IMDb Movies Extensive Dataset*. 2016. <https://kaggle.com/stefanoleone992/imdb-extensive-dataset>.
- Kalra, et al., D. 2022. *Trustworthiness of Indian Youtubers*. Kaggle. <https://doi.org/10.34740/KAGGLE/DSV/4426566>.
- Keating, D., N. Ahmed, F. Nirappil, Stanley-Becker I., and L. Bernstein. 2021. “Coronavirus Infections Dropping Where People Are Vaccinated, Rising Where They Are Not, Post Analysis Finds.” *Washington Post*. <https://www.washingtonpost.com/health/2021/06/14/covid-cases-vaccination-rates/>.
- Laeng, Mathisen, B. 2007. “Why Do Blue-Eyed Men Prefer Women with the Same Eye Color?” *Behavioral*

Ecology and Sociobiology 61(3).

- Levin, D. T. 2000. "Race as a Visual Feature: Using Visual Search and Perceptual Discrimination Tasks to Understand Face Categories and the Cross-Race Recognition Deficit." *Journal of Experimental Psychology* 129(4).
- LUETKEMEIER, et al., M. 2017. "Skin Tattoos Alter Sweat Rate and Na⁺ Concentration." *Medicine and Science in Sports and Exercise* 49(7).
- Madden, et al, J. 2020. "Ready Student One: Exploring the Predictors of Student Learning in Virtual Reality." *PLoS ONE* 15(3).
- Miller, G. A. 1956. "The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information." *Psychological Review* 63(2).
- Moquin, W., and C. Van Doren. 1973. *Great Documents in American Indian History*. Praeger.
- More Americans Are Joining the 'Cashless' Economy*. 2022. <https://www.pewresearch.org/short-reads/2022/10/05/more-americans-are-joining-the-cashless-economy/>.
- National Weather Service Corporate Image Web Team. n.d. *National Weather Service – NWS Billings*. <https://w2.weather.gov/climate/xmacis.php?wfo=byz>.
- O'Brien, Lynch, H. D. 2019. "Crocodylian Head Width Allometry and Phylogenetic Prediction of Body Size in Extinct Crocodyliforms." *Integrative Organismal Biology* 1.
- Ocean Temperature and Salinity Study*. n.d. <https://calcofi.org/>.
- Older People Who Get Covid Are at Increased Risk of Getting Shingles*. 2022. <https://www.washingtonpost.com/health/2022/04/19/shingles-and-covid-over-50/>.
- Physician's Health Study*. n.d. <https://phs.bwh.harvard.edu/>.
- Porath, Erez, C. 2017. "Does Rudeness Really Matter? The Effects of Rudeness on Task Performance and Helpfulness." *Academy of Management Journal* 50.
- Quinn, G. E., C. H. Shin, M. G. Maguire, and R. A. Stone. 1999. "Myopia and Ambient Lighting at Night." *Nature* 399 (6732): 113–14. <https://doi.org/10.1038/20094>.
- Ramachandran, V. 2007. *3 Clues to Understanding Your Brain*. https://www.ted.com/talks/vs_ramachandran_3_clues_to_understanding_your_brain.
- Rates of Laboratory-Confirmed COVID-19 Hospitalizations by Vaccination Status*. 2021. CDC. <https://covid.cdc.gov/covid-data-tracker/#covidnet-hospitalizations-vaccination>.
- Richardson, T., and R. T. Gilman. 2019. "Left-Handedness Is Associated with Greater Fighting Success in Humans." *Scientific Reports* 9 (1): 15402. <https://doi.org/10.1038/s41598-019-51975-3>.
- Stephens, R., and O. Robertson. 2020. "Swearing as a Response to Pain: Assessing Hypoalgesic Effects of Novel "Swear" Words." *Frontiers in Psychology* 11: 643–62.
- Stewart, E. H., B. Davis, B. L. Clemans-Taylor, B. Littenberg, C. A. Estrada, and R. M. Centor. 2014. *Rapid Antigen Group a Streptococcus Test to Diagnose Pharyngitis: A Systematic Review and Meta-Analysis*. 9

(11). <https://doi.org/10.1371/journal.pone.0111727>.

Stroop, J. R. 1935. "Studies of Interference in Serial Verbal Reactions." *Journal of Experimental Psychology* 18: 643–62.

Subach, et al, A. 2022. "Foraging Behaviour, Habitat Use and Population Size of the Desert Horned Viper in the Negev Desert." *Soc.Open Sci* 9.

Sulheim, S., A. Ekeland, I. Holme, and R. Bahr. 2017. *Helmet Use and Risk of Head Injuries in Alpine Skiers and Snowboarders: Changes After an Interval of One Decade*. 51 (1): 44–50. <https://doi.org/10.1136/bjsports-2015-095798>.

Titanic. n.d. <http://www.encyclopedia-titanica.org>.

US COVID-19 Vaccine Tracker: See Your State's Progress. 2021. Mayo Clinic. <https://www.mayoclinic.org/coronavirus-covid-19/vaccine-tracker>.

US Environmental Protection Agency. n.d. *Air Data – Daily Air Quality Tracker*. <https://www.epa.gov/outdoor-air-quality-data/air-data-daily-air-quality-tracker>.

Wahlstrom, et al, K. 2014. "Examining the Impact of Later School Start Times on the Health and Academic Performance of High School Students: A Multi-Site Study." *Center for Applied Research and Educational Improvement*.

Watson, et al., N. 2015. "Recommended Amount of Sleep for a Healthy Adult: A Joint Consensus Statement of the American Academy of Sleep Medicine and Sleep Research Society." *Sleep* 38(6).

Weiss, R. D. 1988. "Relapse to Cocaine Abuse After Initiating Desipramine Treatment." *JAMA* 260(17).

Welcome to the Navajo Nation Government: Official Site of the Navajo Nation. 2011.Retrieved from <https://www.navajo-nsn.gov/>.

Wilson, Woodruff, J. P. 2016. "Vertebral Adaptations to Large Body Size in Theropod Dinosaurs." *PLoS ONE* 11(7).